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Grade	0.00 out of 150.00 (0%)

Question 1

Not answered

Marked out of 3.00

Maladaptive defences

For each of the scenarios below choose the most prominent defence mechanisms used.

Akunna is a 27-year old male university student from southern Nigeria. He was declared missing for 7 days by his roommates in Birmingham, where he was studying physiotherapy. He was later seen in Dundee, picked up from a bus station by the police. He appeared fearful and had no knowledge of how he made the journey to Dundee. He also denied memories of his whereabouts over the past 7 days. He appeared unkempt and exhausted but fully alert and had no evidence of any substance use. His roommates provided a history of severe academic pressures, and issues with the patient's scholarship being cancelled imminently. A fugue state was suspected. (ONE)

Ms A was admitted for depression and suicidal thoughts. She was seen repeatedly clutching to a stuffed animal, and banging her head on her pillow. Whenever a psychiatric interview was attempted, she started crying uncontrollably and pleaded to be sent home. (ONE)

Harmony, a 14-year-old girl, was initially evaluated after an ankle injury during a gymnastic competition at school. After 4 weeks, she again presented with new symptoms of tonic- and clonic-like movements lasting up to an hour – followed by a prolonged inability to move both her legs. She also had trouble recognising some of her school friends. On occasions, especially when her mother was visiting her, she experienced distributed twitching of several body parts. Her mother provided a history of a recent escalation in marital conflicts and Harmony had stopped talking to her mother 5 months before these events. Several video EEG were obtained and she was diagnosed to have a Somatic Symptoms Disorder with attacks and seizures, also known as psychogenic non-epileptic seizure disorder. (ONE)

- Denigration
- Omnipotence
- Dissociation
- Sublimation
- Splitting
- Rationalisation
- Reaction Formation
- Introjection
- Denial
- Conversion
- Suppresion
- Isolation
- Idealisation
- Regresion

Your answer is incorrect.

Explanation:

Akunna is in a fugue state. **Dissociation** is the predominant defence seen in dissociative fugue and amnesia. This involves temporarily but drastically modifying one's sense of personal identity or mental functions to avoid emotional distress. Dissociative amnesia is seen as a response to trauma or uncertainty, with the inability to remember the past being a psychogenic form of coping with the stressor. When explaining fugue and amnesia, invoking dissociation is more popular than the Freudian concept of repression (Mangiulli et al., 2021).

The behaviour exhibited by the patient Ms. A in this question is better explained by regression, rather than introjection, though both are mechanisms invoked in psychodynamic theories of depression. According to Freud, regression causes a reversion to an earlier stage of development (instead of handling unacceptable impulses in a more adult manner). Regression may explain the withdrawal and tearfulness seen in depression. While it is normal for children to manifest regressive acts to communicate their distress, in adults, reverting to earlier stages of development, anticipating being rescued by parents, can often be a maladaptive mechanism. Regressive behaviour is especially problematic in a hospital, as it prolongs hospital stays in the presence of depression (Lokko et al., 2015).

Introjection occurs when our perceptions of significant figures in our lives are internalised where they form part of the structure of the personality (e.g. someone who was raised by a hostile and critical father may themselves feel persecuted by the introjection of this object but also may 'become like' this object at other times). Freud's theory on depression suggests that it is caused by introjection of the aspects of others that make the depressed patient feel anger; leading to 'anger turned inwards'.

Harmony has a conversion disorder (psychogenic non-epileptic seizures). Conversion refers to the switching of mental conflicts into somatic symptoms as in this case. Patients with psychogenic non-epileptic seizures are more likely to have a history of lifetime traumatic experiences, and maladaptive defences such as conversion and dissociative tendencies with lower resilience than their peers (Masi et al., 2020).

Ref: Mangiulli I, et al. A critical review of case studies on dissociative amnesia. Clinical Psychological Science. 2021 Jun 8:21677026211018194.

Lokko HN, Stern TA. Regression: Diagnosis, evaluation, and management. The primary care companion for CNS disorders. 2015 May 14;17(3):27221.

Masi G, et al. Psychopathological features in referred adolescents with psychogenic nonepileptic seizures with or without epilepsy. Epilepsy Behav 2020; 112: 107431.

The correct answer is:
Akunna is a 27-year old male university student from southern Nigeria. He was declared missing for 7 days by his roommates in Birmingham, where he was studying physiotherapy. He was later seen in Dundee, picked up from a bus station by the police. He appeared fearful and had no knowledge of how he made the journey to Dundee. He also denied memories of his whereabouts over the past 7 days. He appeared unkempt and exhausted but fully alert and had no evidence of any substance use. His roommates provided a history of severe academic pressures, and issues with the patient's scholarship being cancelled imminently. A fugue state was suspected. (ONE) → Dissociation, Ms A was admitted for depression and suicidal thoughts. She was seen repeatedly clutching to a stuffed animal, and banging her head on her pillow. Whenever a psychiatric interview was attempted, she started crying uncontrollably and pleaded to be sent home. (ONE) → Regression,

Harmony, a 14-year-old girl, was initially evaluated after an ankle injury during a gymnastic competition at school. After 4 weeks, she again presented with new symptoms of tonic- and clonic-like movements lasting up to an hour – followed by a prolonged inability to move both her legs. She also had trouble recognising some of her school friends. On occasions, especially when her mother was visiting her, she experienced distributed twitching of several body parts. Her mother provided a history of a recent escalation in marital conflicts and Harmony had stopped talking to her mother 5 months before these events. Several video EEG were obtained and she was diagnosed to have a Somatic Symptoms Disorder with attacks and seizures, also known as psychogenic non-epileptic seizure disorder. (ONE) → Conversion

Question 2

Not answered

Marked out of 4.00

Delusions in clinical practice

Which one of the following descriptions below correctly refers to the psychopathological terms above?

A 34-year-old psychotic patient sees two white cars pulling up in front of his home. He is convinced that he is about to be accused of being a paedophile.

A 40-year-old psychotic patient who believes that parts of the Bible are about him directly.

A 33-year-old man requested admission to a hospital because he had become convinced that God had granted him the greatest possible sort of mind and that coming into contact with him would cure others of mental illnesses.

A 34-year-old man with psychotic patient on the ward recalled being anxious and a sense of 'something about to happen' prior to his admission.

Delusional mood
Delusional jealousy
Autochthonous delusion
Delusions of reference
Delusional memory
Delusions of grandiosity
Delusions of infestation
Delusional perception
Delusional elaboration
Delusions of control

Your answer is incorrect.

Explanation:

A delusional perception is a primary delusion which is recalled as having arisen as a result of a perception (e.g. a patient who, on seeing two white cars pull up in front of his house, became convinced that he was therefore about to be wrongly accused of being a paedophile). The percept is a real external object, not a hallucinatory experience.

A delusion of reference is a delusional belief that external events or situations have been arranged in such a way as to have particular significance for, or to convey a message to, the affected individual. The patient may believe that television news items are referring to them or that parts of the Bible are about them directly.

A grandiose delusion is a delusional belief that one has special powers or is unusually rich or powerful, or that one has an exceptional destiny (e.g. a man who requested admission to hospital because he had become convinced that God had granted him 'the greatest possible sort of mind' and that coming into contact with him would cure others of mental illnesses.)

Delusional mood is recalled as arising following a period when there is an abnormal mood state characterised by anticipatory anxiety, a sense of 'something is about to happen' and an increased sense of the significance of minor events. The development of the formed delusion may come as a relief to the patient in this situation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 107, 112.

The correct answer is: A 34-year-old psychotic patient sees two white cars pulling up in front of his home. He is convinced that he is about to be accused of being a paedophile. → Delusional perception, A 40-year-old psychotic patient who believes that parts of the Bible are about him directly. → Delusions of reference, A 33-year-old man requested admission to a hospital because he had become convinced that God had granted him the greatest possible sort of mind and that coming into contact with him would cure others of mental illnesses. → Delusions of grandiosity, A 34-year-old man with psychotic patient on the ward recalled being anxious and a sense of 'something about to happen' prior to his admission. → Delusional mood

Question 3

Not answered

Marked out of 4.00

MoK Assessment CXEMI014

Rating scales in clinical use

Identify the relevant rating scales for each of the following conditions

1. Choose TWO observer-rated tests to monitor depression in a woman recently diagnosed with postnatal depression (Choose Two)

2. Choose the most appropriate tool to screen for depression in mothers following childbirth

3. Choose one scale to assess EPSEs due to antipsychotic use.

Scale for Assessment of Positive Symptoms
Brief Psychiatric Rating scale
Present state examination
Beck's depression inventory
Edinburgh postnatal depression scale
General Health questionaire
Hamilton Depression scale
Montgomery Asberg Depression Rating Scale
Simpson Angus Scale

Explanation:

The Hamilton Depression Rating Scale is a widely used scale for measuring depression, consisting of a list of 17 items, designed to be rated by a clinician on three-point and five-point scales. The symptoms are: depressed mood, guilt feelings, suicide, insomnia (initial or early), insomnia (middle), insomnia (terminal or late), disturbed work and activities, psychomotor retardation, psychomotor agitation, anxiety (psychological), anxiety (somatic), gastrointestinal symptoms, somatic symptoms, sexual dysfunction or menstrual disturbance, hypochondriasis, weight loss, and lack of insight.

The Montgomery-Asberg Depression Rating Scale (MADRS) is a 10-item observer-rated scale of depression. Each item is rated 0-6, with a total score obtained.

The Edinburgh Postnatal Depression Scale is a 10-item tool developed for screening postpartum women for depression in outpatient clinics, home visits, or during postpartum appointments. It contains dimensions of depression, anxiety, anhedonia, and self-harm/suicide.

The Simpson-Angus Scale is a clinical tool commonly used to detect and monitor extrapyramidal symptoms. It rates 10 items from 0 to 4 points, yielding a total body score divided by 10, and considered normal up to 0.3.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 97.

Smith-Nielsen J et al. Validation of the Edinburgh Postnatal Depression Scale against both DSM-5 and ICD-10 diagnostic criteria for depression. BMC psychiatry. 2018 Dec;18(1):1-2.

Blanchet PJ, Rompré PH. Clinimetric evaluation of the Simpson-Angus Scale in older adults with schizophrenia. Journal of clinical psychopharmacology. 2014 Feb 1;34(1):36-9.

The correct answer is:
Choose TWO observer-rated tests to monitor depression in a woman recently diagnosed with postnatal depression (Choose Two) - Hamilton Depression scale, Montgomery Asberg Depression Rating Scale

Choose the most appropriate tool to screen for depression in mothers following childbirth - Edinburgh postnatal depression scale

Choose one scale to assess EPSEs due to antipsychotic use - Simpson-Angus Scale

Question 4

Not answered

Marked out of 5.00

Disorders of perception

Which of the listed descriptive psychopathological terms refers to the following symptoms?

A 34-year-old gentleman in the psychiatric outpatient clinic mentioned 'Just as I was about to fall asleep, I heard a voice calling my name which then startled me back to wakefulness only to find no one there.'

A 20-year-old university student mentioned that while she was walking across a lonely park at night; she became anxious on mistaking bushes rustling in the wind as a hidden attacker but later calmed down.

A woman with bilateral cataracts says that she sees a number of her dead relatives walking in the house.

A 46-year-old woman on the psychiatric ward says 'All my blood has dropped into my legs, and I can feel a water level in my chest.'

A 37-year-old gentleman on the psychiatric ward described 'I thought my life was outside my feet and made them vibrate.'

Charles Bonnet Syndrome
Kinaesthetic hallucination
Hypnopompic hallucination
Reflex Hallucination
Haptic Hallucinations
Pareidolic illusion
Affect illusions
Hygric hallucinations
Hypnagogic hallucination
Compete illusion

Choose...

Your answer is incorrect.

Explanation:

A hypnagogic hallucination is a transient false perception experienced while on the verge of falling asleep. These may have the characteristics of true or pseudo-hallucinations and are most commonly visual or auditory. While they are sometimes seen in narcolepsy or affective illnesses, they are not indicative of ill health and are frequently reported by healthy people.

An illusion is a type of false perception in which the perception of a real-world object is combined with internal imagery to produce a false internal percept. Three types are recognised: affect, completion, and pareidolic illusions. In affect illusion, there is a combination of a heightened emotion and misperception. Completion illusions rely on our brain's tendency to 'fill in' presumed missing parts of an object to produce a meaningful percept and are the basis for many types of optical illusions. Both these types of illusions resolve on closer attention. Pareidolic illusions are meaningful percepts produced when experiencing a poorly defined stimulus (e.g. seeing faces in a fire or clouds).

Charles Bonnet syndrome (phantom visual images) is a condition in which complex visual hallucinations occur in the absence of any psychopathology and in clear consciousness. It is associated with either central or peripheral reduction in vision. The images may be static or in motion. They may be enjoyable or distressing. Patients may retain insight.

A hygric hallucination is a tactile hallucination of feeling fluid (such as water running from the head into the stomach).

Kinaesthetic hallucinations affect the muscles and joints, and the patient feels that their limbs are being twisted, pulled, or moved. Vestibular sensations such as sinking in the bed or flying through the air can also be hallucinated and are best regarded as a variant of kinaesthetic hallucinations.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 112-113.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. pp. 29, 34, 124.

The correct answer is: A 34-year-old gentleman in the psychiatric outpatient clinic mentioned 'Just as I was about to fall asleep, I heard a voice calling my name which then startled me back to wakefulness only to find no one there.' → Hypnagogic hallucination, A 20-year-old university student mentioned that while she was walking across a lonely park at night; she became anxious on mistaking bushes rustling in the wind as a hidden attacker but later calmed down. → Affect illusions, A woman with bilateral cataracts says that she sees a number of her dead relatives walking in the house. → Charles Bonnet syndrome, A 46-year-old woman on the psychiatric ward says 'All my blood has dropped into my legs, and I can feel a water level in my chest.' → Hygric hallucinations, A 37-year-old gentleman on the psychiatric ward described 'I thought my life was outside my feet and made them vibrate.' → Kinaesthetic hallucination

Question 5

Not answered

Marked out of 5.00

MoK Assessment CXEMI026

Conflicts and Defences

Choose the best option from the list above for each description below.

1. Which defence mechanisms are used in patients with obsessive-compulsive disorder (Choose three answers)

2. Which defence mechanism is used when a person tries to negate his actions?

3. Which defence mechanism is used in patients with impulse control disorder?

- Acting out
- Denial
- Displacement
- Dissociation
- Undoing
- Introjection
- Isolation
- Passive-aggressive
- Projection
- Reaction formation

Explanation:

Freud coined the term 'obsessional neurosis' to describe the present day concept of OCD. He considered obsessional neurosis to be the result of regression from the Oedipal state to the pre-genital anal-erotic stage of development as a defence against aggressive or sexual (unconscious) impulses. Associated defences are isolation (obsessional thoughts), undoing (compulsive rituals) and reaction formation. Symptoms occur when these defences fail to contain the anxiety.

Isolation (of affect) is a defence mechanism in which emotion is detached from an idea and rendered unconscious, leaving the idea bland and emotionally flat. The mechanisms by which the detachment is effected include repetitive intrusive thoughts and ritualistic behaviour.

Undoing: When a person acts inappropriately, it sometimes produces anxiety. To counter this anxiety, the person may try to negate the original behaviour. For example, a child who becomes unruly at the dinner table but then offers to help during cleanup.

Reaction Formation: When someone is confronted with disturbing desires or impulses, he or she may actively express the opposite impulse. This involves openly displaying a particular attitude that is opposite of disturbing repressed traits.

Seeing, hearing or doing, but refusing to acknowledge what one sees, hears or does, and negating these are examples of **denial**. Denial avoids becoming aware of some painful aspect of reality.

In **acting out**, patients directly express unconscious wishes or conflicts through action to avoid being conscious of either the accompanying idea or the affect.The unconscious fantasy is lived out impulsively in behaviour, thereby gratifying the impulse instead of prohibiting it, seen in impulse control disorders. Tantrums, apparently motiveless assaults, and pleasureless promiscuity are also typical examples.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 385.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 577, 1066.

Isolation of affect. *Oxford Reference.* Retrieved 4 Mar. 2022, from https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100012916.

The correct answer is:
Which defence mechanisms are used in patients with obsessive-compulsive disorder? (Choose three answers) - Undoing, Isolation, Reaction formation

Which defence mechanism is used when a person tries to negate his actions? - Denial

Which defence mechanism is used in patients with impulse control disorder? - Acting out

Question 6

Not answered

Marked out of 3.00

Types of Mutations

For the following mutations, select the most appropriate term

Creation of a new stop codon	Choose...
This mutation does not alter the amino acid product	Choose...
Substitution of a purine for a purine	Choose...

In frame
Expansion
Transversion
Nonsense
Frameshift
Missense
Silent
Transition

Your answer is incorrect.

Explanation:

A **nonsense mutation** is any mutation that produces one of the stop codons, causing premature termination of protein synthesis by the gene in which it occurs. In nonsense mutation, the new codon is usually UUA, UGA or UAG, all of which signals 'stop' to the amino acid sequence resulting in a nonfunctional protein.

A **silent mutation** is any mutation that has no effect on the phenotype, either because the altered codon happens to specify the same amino acid as the original one, or because it occurs in an intron that has no genetic function. Thus, a silent mutation causes no change in protein product.

A point mutation is a mutation involving the alteration of a single base pair in a segment of DNA, whereby substitutions of one base with another occurs. (**Transition** = purine replaced by another purine or pyrimidine replaced by another pyrimidine (e.g. A to G); Transversion = purine replaced by a pyrimidine or vice versa (e.g. A to T)).

In missense mutation, the new mutant codon specifies a different amino acid with variable effects on final protein product. For example, haemophilia, sickle cell anaemia. In frame shift mutations, the deletion or insertion is not in multiples of three codons e.g. a segment of 5 bases deletion mutations.

For a short illustrative description see

The correct answer is: Creation of a new stop codon → Nonsense, This mutation does not alter the amino acid product → Silent, Substitution of a purine for a purine → Transition

Question 7

Not answered

Marked out of 6.00

MoK Neurosciences CXEMI012

Cerebral circulation disorders

Identify 3 common symptoms seen in following types of cerebral circulation disorders.

1. Carotid system TIA ✖ ✖ ✖

2. Vertebrobasilar system TIA ✖ ✖

Dysarthria
Amaurosis Fugax
Hemiparesis
Ataxia
Horner syndrome
Diplopia
Nystagmus
Myoclonus
Aphasia
Agraphria

Explanation

Carotid system TIA is also called as anterior circulation TIA. It is characterised by amaurosis fugax, aphasia, hemiparesis, hemisensory loss and hemianopic visual loss due to the involvement of watershed or highly oxygen dependent cortical regions.

Vertebrobasilar system TIA is also called as posterior circulation TIA. It affects brain stem cranial nerve nuclei leading to diplopia, vertigo, vomiting, choking, dysarthria, and ataxia, alexia without agraphia, hemisensory loss, hemianopic visual loss, transient global amnesia, quadriparesis and loss of consciousness (rare).

The correct answer is:

Carotid system TIA - Amaurosis fugax, Hemiparesis, Aphasia,

Vertebrobasilar system TIA - Dysarthria, Ataxia, Diplopia

Question 8

Not answered

Marked out of 5.00

Speech disorder

Identify the correct terms that represent each of the following speech disorders

Akinetic mutism
Syntactical aphasia
Logoclonia
Stammering
Logorrhoea
Alogia
Selective mutism
Palialia
Hysterical mutism
Normal apahsia

Decrease in spontaneous talking occurring in depression	C
Increased quantity of speech seen in early manic states	C
The patient is mute but remain aware of the environment though cannot move or respond	C
Difficulty in finding the correct name for an object	C
Often seen in children who refuse to speak to certain people	Choose...

Your answer is incorrect.

Explanation:

Alogia is poverty of thoughts as observed by the absence of spontaneous speech. It is a negative symptom of schizophrenia and a symptom of depressive illness.

Logorrhea is excess speech or 'verbal diarrhoea'. It is a symptom of mania.

In **akinetic mutism**, which is associated with lesions of the upper midbrain or posterior diencephalon, there is mutism, and the patient appears to be aware of the environment.

In **nominal aphasia** the patient cannot name objects, although they have plenty of words at their disposal. Difficulty in finding the correct word can occur in other varieties of aphasia, but in nominal aphasia it is the outstanding disorder. Nominal aphasia may be found with either diffuse brain damage or with focal lesions involving the dominant temporoparietal region (e.g. angular gyrus).

Selective mutism may occur in children who refuse to speak to certain people; for example, the child may be mute at school but speak at home.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 102, 113.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. pp. 56-59.

The correct answer is: Decrease in spontaneous talking occurring in depression → Alogia, Increased quantity of speech seen in early manic states → Logorrhoea, The patient is mute but remain aware of the environment though cannot move or respond → Akinetic mutism, Difficulty in finding the correct name for an object → Nominal aphasia, Often seen in children who refuse to speak to certain people → Selective mutism

Question 9

Not answered

Marked out of 3.00

Chromosomal loci

Select the most appropriate chromosomal locus for each of the scenario given below

A 7-year-old girl has a large mandible and an open-mouthed expression which reveals the tongue. She has severe intellectual disability, with poor communication skills and little or absent speech. She has an unusual manner of laughing when stimulated.

A 3-year-old girl repeatedly presents with respiratory infections. On examination, her larynx appears to be maldeveloped, and she has an unusually small head with a round face and a small chin with folds of skin over her eyes. The mother recalls a distinctive crying sound that became less noticeable after age 2.

A 4-year-old girl is brought to the A&E with severe breathing problems. On examination, there is excessive drooling, with reduced muscle tone in her limbs. Developmental history is normal up to the age of 18 months apart from repeated problems related to gastroesophageal reflux. Growth charts reveal a slowing of head growth beginning at approximately 5 - 6 months of age.

Chromosome 5
Chromosome 7
Chromosome 4
Chromosome 15
Chromosome Y
Chromosome X
Chromosome 11
Chromosome 1
Chromosome 21
Chromosome 22
Mitochondria DNA

Your answer is incorrect.

Explanation:

Angelman ('happy puppet') syndrome (Case 1):

Microdeletion 15q11-13 of **maternal origin**; 80% due to deletion of maternally derived chromosome 15

Incidence: 1:10,000

Clinical features: happy disposition, paroxysmal laughter, absence of speech, hand flapping, clapping, ataxia (jerky limb movements, gait problems)

Severe/profound ID

Facial features: fair hair, blue eyes, microcephaly, flattened occiput, long face, prominent jaw, wide mouth, widely-spaced teeth, thin upper lip, and pointed chin

Epilepsy (90%); EEG is highly characteristic with changes noted as early as age 3

Cri-du-chat (Case 2):

Partial monosomy; karyotype 5p- (varies from deletion of a small band at 5p15.2 to entire arm of 5p)

About 85% of the deletions arise **spontaneously** and the majority are of **paternal** origin

Incidence: 1:35,000

More common in **females** (4:3)

Infant has a characteristic **high-pitched cry** that resembles a cat

Hyperactivity is a problem for a substantial proportion of children, but may improve with age

Rett syndrome (Case 3):

Believed to be caused by **dominant X-linked gene**

Degenerative disorder that almost exclusively affects **females**

Initially normal development

Between ages 1- 4 years of age: purposeful hand movements and spoken language are lost; stereotypical hand movements (**hand-wringing**) and autistic symptoms are seen

Ataxia, facial grimacing, teeth-grinding, and loss of speech

Irregular respiration with episodes of hyperventilation, apnea, and breath-holding

Progressive **gait disturbance, scoliosis,** and **seizures**

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 808, 815.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 103.

The correct answer is: A 7-year-old girl has a large mandible and an open-mouthed expression which reveals the tongue. She has severe intellectual disability, with poor communication skills and little or absent speech. She has an unusual manner of laughing when stimulated. → Chromosome 15, A 3-year-old girl repeatedly presents with respiratory infections. On examination, her larynx appears to be maldeveloped, and she has an unusually small head with a round face and a small chin with folds of skin over her eyes. The mother recalls a distinctive crying sound that became less noticeable after age 2. → Chromosome 5, A 4-year-old girl is brought to the A&E with severe breathing problems. On examination, there is excessive drooling, with reduced muscle tone in her limbs. Developmental history is normal up to the age of 18 months apart from repeated problems related to gastroesophageal reflux. Growth charts reveal a slowing of head growth beginning at approximately 5 - 6 months of age. → Chromosome X

Question 10

Not answered

Marked out of 4.00

Personality tests

For each of the following tests given below, choose their correct names from the list given:

This test measures personality along three dimensions: extraversion, neuroticism, and psychoticism	Eysenck Personality Questionnaire
It is one of the objective personality tests that includes hypochondriasis as a personality factor	Minnesota multiphasic personality inventory
A projective personality test with pictures of ambiguous persons and scenes to make up a story	Thematic apperception test
This test is useful to assess premorbid intelligence	National adult reading test

Stroop test
Wisconsin card sorting test
Trail making test
Raven's progressive matrices
Eysenck Personality Questionnaire
Rorschach inkblot test
Thematic apperception test
National adult reading test
Paired association test
Rey Osterreith Test
Minnesota multiphasic personality inventory

Your answer is incorrect.

Explanation:

Eysenck's Personality Questionnaire is a revised version of the Eysenck Personality Inventory, measuring three major dimensions of personality - namely extraversion, neuroticism, and psychoticism - and including a lie scale.

The Minnesota Multiphasic Personality Inventory (MMPI) is a self-report questionnaire consisting of 567 questions covering eight areas of psychopathology and two areas of personality type. The scales are: Hypochondriasis, Depression, Hysteria, Psychopathic deviate, Masculinity-femininity, Paranoia, Psychasthenia, Schizophrenia, Hypomania, and Social introversion. There are also four additional validity (test-taking attitude) scales.

The Thematic Apperception Test is a projective personality test. It consists of a series of 10 black-and-white pictures that depict individuals of both sexes and of different age groups, who are involved in a variety of different activities. The stories that the patient makes up concerning the pictures, according to the projective hypothesis, reflect the patient's own needs, thoughts, feelings, stresses, wishes, desires, and views of the future.

First published in 1982, the 50-item National Adult Reading Test (NART) is a widely adopted method for estimating premorbid intelligence both for clinical and research purposes.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 96.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 59-60.

Bright P et al. The National Adult Reading Test: restandardisation against the Wechsler adult intelligence scale—fourth edition. Neuropsychological Rehabilitation. 2018 Aug 18;28(6):1019-27.

The correct answer is: This test measures personality along three dimensions: extraversion, neuroticism, and psychoticism → Eysenck Personality Questionnaire, It is one of the objective personality tests that includes hypochondriasis as a personality factor → Minnesota multiphasic personality inventory, A projective personality test with pictures of ambiguous persons and scenes to make up a story → Thematic apperception test, This test is useful to assess premorbid intelligence → National adult reading test

Question 11

Not answered

Marked out of 4.00

Neuropathological features

You are working in a neuroscience department, and you have a special interest in dementias. For each of the definitions given below, identify the neuropathological terms used to describe them.

Weakly eosinophilic spherical cytoplasmic inclusions

This protein aggregates to form oligomers that turn into Lewy bodies

Ballooned or swollen cortical neurons, with loss of Nissl's substance, oval in shape with peripherally displaced nucleus

Rounded, homogeneous neuronal inclusions, faintly visible on H&E staining, containing 3Rtau, and strongly argyrophilic with the Bielschowsky technique

Microvacuolation
Prion proteins
Spongiform encephalopathy
Lewy bodies
Alpha-synuclein
Pick bodies
Senile plaques
Pick cells
Hirano bodies

Your answer is incorrect.

Explanation:

Lewy bodies are neuronal cytoplasmic inclusions. Typical (brainstem) Lewy bodies are roughly spherical with an eosinophilic core surrounded by a paler 'halo'.

Alpha-synucleinopathies are disorders in which there is accumulation of the membrane-associated, presynaptic, protein alpha-synuclein. Aggregates may be visible on routine sections; many more are revealed by immunohistochemistry. This group includes Parkinson disease, other Lewy body diseases, and multiple-system atrophy. In Lewy body dementia, for unknown reasons, alpha-synuclein proteins aggregate to form oligomers, eventually turning into 'Lewy bodies' and Lewy neurites, as neurons degenerate.

Pick disease is characterised by the presence of Pick bodies: rounded, homogeneous neuronal cytoplasmic inclusions, faintly visible on H&E staining. Immunohistochemical staining shows that these inclusions contain 3Rtau. Pick bodies are also strongly argyrophilic with the Bielschowsky technique. Ballooned neurons (sometimes referred to as Pick cells) are also frequent. Pick cells are swollen cortical neurons that are a characteristic, but nonspecific, feature of Pick disease.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. pp. 6, 187, 192-193.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 493.

The correct answer is: Weakly eosinophilic spherical cytoplasmic inclusions → Lewy bodies, This protein aggregates to form oligomers that turn into Lewy bodies → Alpha-synuclein, Ballooned or swollen cortical neurons, with loss of Nissl's substance, oval in shape with peripherally displaced nucleus → Pick cells, Rounded, homogeneous neuronal inclusions, faintly visible on H&E staining, containing 3Rtau, and strongly argyrophilic with the Bielschowsky technique → Pick bodies

Question 12

Not answered

Marked out of 3.00

Less reported adverse effects

For each of the adverse effects given below, identify the most likely drug associated with it.

Severe sweating	Choose...
Torsades de pointes	Choose...
Allergic dermatitis and photosensitivity	Choose...

Carbamazepine
Droperidol
Venlafaxine
Lorazepam
Chlorpromazine
Lamotrigine
Mirtazapine
Lithium
Bupropion
Acamprosate
Sildenafil

Your answer is incorrect.

Explanation:

Adrenergic agents are implicated in causing antidepressant-induced **sweating**, with adrenergic antidepressants like duloxetine, **venlafaxine** and reboxetine having a higher propensity. All antidepressants may induce excessive sweating, though some drugs reportedly have weaker association (fluvoxamine, bupropion and vortioxetine). SSRIs have a comparable risk, not significantly different from SNRIs. Among SSRIs, paroxetine and sertraline have a higher risk than others. Both alpha blockers (e.g. clonidine, terazosin) and beta blockers can improve antidepressant-induced sweating by blocking adrenergic output. The sympathetic postganglionic innervation of sweat glands can also be blocked with anticholinergics like benztropine or glycopyrrolate. Serotonergic antagonism with cyproheptadine has also been suggested as a treatment option. Antidepressant induced sweating is estimated to occur in 5-14% of patients. This is not a dose dependent effect.

Several antipsychotic drugs, belonging both to typical and atypical classes (e.g. sertindole, thioridazine, **droperidol**, and haloperidol) are known to be associated with an increased risk of QTc prolongation, potentially leading to serious ventricular arrhythmias such as **torsades de pointes** and sudden cardiac death. In 1998, droperidol was withdrawn from the market because of its torsadogenic potential and in 2005, thioridazine was withdrawn from the global market because of the risk of sudden cardiac death. Droperidol has since become available again and is seeing a resurgence of use in some countries (its initial withdrawal was voluntary, so reintroduction is not prohibited).

Chlorpromazine has the potential to cause skin **photosensitivity** and patients should avoid undue exposure to sunlight. Allergic dermatitis can also occur in short term with chlorpromazine. Long-term chlorpromazine use can cause blue-gray discolouration of skin areas exposed to sunlight. This may be reversible.

References:

Beyer C, Cappetta K, Johnson JA, et al. Meta-analysis: risk of hyperhidrosis with second-generation antidepressants. Depress Anxiety. 2017;34:1134-1146.

Oteri A et al. Prescribing pattern of antipsychotic drugs during the years 1996–2010: a population-based database study in Europe with a focus on torsadogenic drugs. British journal of clinical pharmacology. 2016 Aug;82(2):487-97.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 49.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 156.

The correct answer is: Severe sweating → Venlafaxine, Torsades de pointes → Droperidol, Allergic dermatitis and photosensitivity → Chlorpromazine

Question 13

Not answered

Marked out of 3.00

Plasma steady state

For each of the following drugs, identify the time taken to achieve steady state plasma levels

Lithium	Choose...
Clozapine	Choose...
Sodium valproate	Choose...

6 weeks
4 weeks
1 day
2 weeks
2-3days
2 months
3 months
5 days

Your answer is incorrect.

Explanation:

Lithium reaches steady state levels in 5 days. Clozapine and valproate both reach steady state levels in 2-3 days.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 835-836.

The correct answer is: Lithium → 5 days, Clozapine → 2-3 days, Sodium valproate → 2-3 days

Question 14

Not answered

Marked out of 1.00

A sudden fright or shock is usually the precipitant for which culture-related syndrome?

Select one:

- Koro
- Amok
- Dhat syndrome
- Latah
- Windigo

Your answer is incorrect.

Explanation:

Latah has been found mainly in Malaysian and Indonesian cultures in which a person - after experiencing a sudden fright or shock - displays abnormal suggestibility accompanied by echolalia, echopraxia, autonomic obedience, and a state resembling a trance. Originally reported as a common phenomenon among young women in Malaysia, it is becoming less prevalent, (particularly in urban areas) and is now found almost exclusively among older women who have known of the phenomenon since they were young.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

Tseng WS. From peculiar psychiatric disorders through culture-bound syndromes to culture-related specific syndromes. Transcultural psychiatry. 2006 Dec;43(4):554-76.

The correct answer is: Latah

Question 15

Not answered

Marked out of 1.00

With respect to vulnerability for depression, poor coping style is best described as which of the following?

Select one:

- Modifying factor
- Maintaining factor
- Precipitating factor
- Resilience factor
- Protective factor

Your answer is incorrect.

Brown et al. identified the following factors as maintaining factors for depression: a. Further negative life events b. Persistent poor quality social support c.Poor coping style characterized by self-blame and helplessness, denial of problems, poor problem-solving ability, blaming others or external forces d. Inability to obtain adequate social support due to fear of intimacy and e. Low educational level.

The correct answer is: Maintaining factor

Question 16

Not answered

Marked out of 1.00

Which of the following is not a central antipsychiatry belief?

Select one:

- Medication and hospitalisation are harmful to the individual so treated
- The scientific method cannot explain the subjective abnormalities of mental disorder as no direct observation can take place
- Mental disorder can be best explained by social, ethical and political factors
- Mind is a bodily organ and can be diseased
- The labelling of individuals as 'ill' is an artificial device used by society to maintain its stability in the face of challenge

Your answer is incorrect.

Explanation:

Central anti-psychiatry beliefs:

1.The mind is not a bodily organ and so cannot be diseased.

2.The scientific method cannot explain the subjective abnormalities of mental disorder, as no direct observation can take place.

3.Mental disorder can best be explained by social, ethical, or political factors.

4.The labelling of individuals as 'ill' is an artificial device used by society to maintain its stability in the face of challenges.

5.Medication and hospitalisation are harmful to the individual so treated.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 28.

The correct answer is: Mind is a bodily organ and can be diseased

Question 17

Not answered

Marked out of 1.00

Which of the following is true with regards to Changing Minds campaign?

Select one:

- Is focussed on geriatric mental health issues
- Is an initiative from World Health Organisation
- Aims to highlight the treatability of mental disorders
- Provides a legal clause against stigma
- Is an ongoing programme against stigma

Your answer is incorrect.

Explanation:

The World Health Organization and the World Psychiatric Association both recognise that the stigma and discrimination associated with mental disorders are strongly linked to suffering, disability, and poverty. Concern about the stigma of mental illness precipitated the Royal College of Psychiatrists' 5-year 'Changing Minds' campaign (1998-2003) with the aim of promoting positive images of mental illness, challenging misrepresentations and discrimination, encouraging patient advocacy, and educating the public about the real nature and treatability of mental disorder.

Luty J et al. Effectiveness of Changing Minds campaign factsheets in reducing stigmatised attitudes towards mental illness. Psychiatric Bulletin. 2007 Oct;31(10):377-81.

The correct answer is: Aims to highlight the treatability of mental disorders

Question 18

Not answered

Marked out of 1.00

The most common reason for doctors to be taken to the court is

Select one:

- Problems in the doctor/patient relationship
- Misuse and abuse
- Negligence
- Not obtaining informed consent
- Breaking confidentiality

Your answer is incorrect.

Negligence: This requires the plaintiff to prove three things 1. That the doctor owed a duty of care to the particular patient 2. That the doctor was in breach of the appropriate standard of care imposed by the law. 3. That the breach in duty of care caused the patient harm, meriting compensation.

The correct answer is: Negligence

Question 19

Not answered

Marked out of 1.00

With respect to population studies, cultural analysis refers to

Select one:

- Racial profile of groups
- Norms in a group
- Beliefs among the group members
- Lingual diversity in a group
- Geographical clusters of separation

Your answer is incorrect.

Cultural analysis refers to the study of norms or behavior patterns that are typical of specific cultural groups. These norms are often passed down from generation to generation by observational learning by the groups gatekeepers - parents, teachers, religious leaders, and peers. Cultural norms include how people choose marriage partners, attitudes toward alcohol consumption, and acceptance (or rejection) of spanking children.

The correct answer is: Norms in a group

Question 20

Not answered

Marked out of 1.00

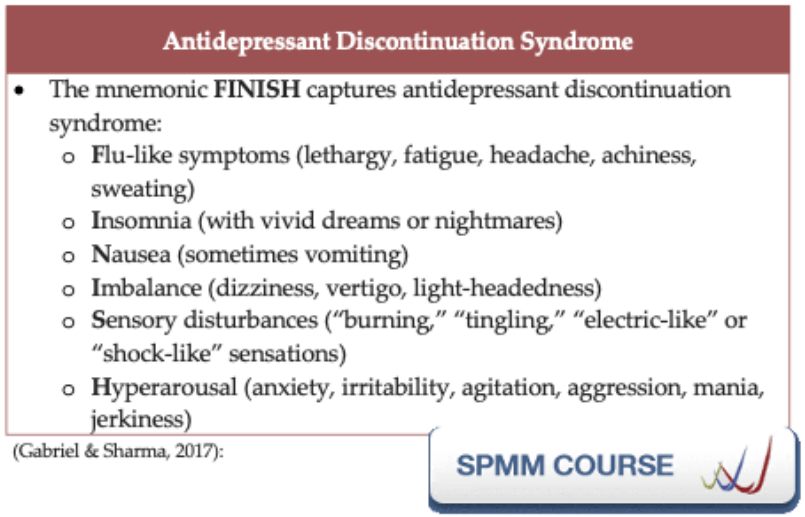
Which one among the following is not a feature of SSRI discontinuation syndrome

Select one:

- Increased dreams
- Dizziness
- Insomnia
- Myoclonic Jerks
- Electric shock like sensations

Your answer is incorrect.

Explanation:



Ref: Gabriel M, Sharma V. Antidepressant discontinuation syndrome. CMAJ. 2017 May 29;189(21):E747.

The correct answer is: Myoclonic Jerks

Question 21

Not answered

Marked out of 1.00

Which of the following form the basis of official social classification in Britain?

Select one:

- Residential status
- Land ownership
- Financial status
- Educational status
- Occupational status

Your answer is incorrect.

Explanation:

In Britain, social class is a derived classification achieved by mapping occupation and employment status to class categories.

Ref: https://sru.soc.surrey.ac.uk/SRU9.html

The correct answer is: Occupational status

Question 22

Not answered

Marked out of 1.00

Mania is often characterized by the presence of

Select one:

- Delusions of control
- Delusions of reference
- Delusions of love
- Delusions of grandiosity
- Delusions of pregnancy

Your answer is incorrect.

Explanation:

Grandiose delusions are most commonly associated with manic psychosis in the context of bipolar affective disorder.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. pp. 51-52.

The correct answer is: Delusions of grandiosity

Question 23

Not answered

Marked out of 1.00

Mr. Y is an 83 year old man seen in the outpatient clinic. He is complaining of vivid dreams. His wife reports that he has been having difficulty with his thinking, though this fluctuates. They no longer sleep in the same bed, as Mr. Y has grabbed his wife in his sleep on a few occasions.

Select one:

- Alzheimer dementia
- Fronto-temporal dementia
- Lewy body dementia
- Chronic subdural haematoma
- Huntington dementia

Your answer is incorrect.

Explanation:

Clinical features of dementia with Lewy bodies:

1.Fluctuations are typically **delirium-like**, occurring as spontaneous alterations in cognition, attention, and arousal

2.Recurrent, complex visual hallucinations (people, children, and animals): up to **80%** of cases

3.Spontaneous **parkinsonian features** (not due to medications or stroke): over **85%**

4.REM sleep behaviour disorder: parasomnia manifested by recurrent **dream enactment behaviour** associated with an **absence of normal REM sleep atonia**; occurs in **76%** of cases

5.Severe antipsychotic sensitivity is **now a supportive**, rather than core, **feature** because reduced prescribing of D2 receptor blocking antipsychotics limits its diagnostic usefulness

6.New supportive features in the criteria: **hypersomnia** (usually excessive daytime sleepiness) and **hyposmia** (occurs earlier in DLB than in AD)

Ref: McKeith IG et al. Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. Neurology. 2017 Jul 4;89(1):88-100.

The correct answer is: Lewy body dementia

Question 24

Not answered

Marked out of 1.00

Which ICD-11 diagnosis incorporates the concept of ICD-10 neurasthenia?

Select one:

- Complex post-traumatic stress disorder
- Bipolar type II disorder
- Body integrity dysphoria
- Body dysmorphic disorder
- Bodily distress disorder

Your answer is incorrect.

Explanation:

ICD-11 disorders of bodily distress and bodily experience encompass two disorders: bodily distress disorder and body integrity dysphoria. ICD-11 bodily distress disorder replaces ICD-10 somatoform disorders and also includes the concept of ICD-10 neurasthenia.

Ref: Reed GM et al. Innovations and changes in the ICD-11 classification of mental, behavioural and neurodevelopmental disorders. World Psychiatry. 2019 Feb;18(1):3-19.

The correct answer is: Bodily distress disorder

Question 25

Not answered

Marked out of 1.00

Excessive doubt and distrust in relationships is a characteristic feature of which of the following personality disorder?

Select one:

- Schizoid
- Antisocial
- Schizotypal
- Borderline
- Paranoid

Your answer is incorrect.

Explanation:

Persons with paranoid personality disorder have long-standing suspiciousness and mistrust of people in general. Clinical features of patients with paranoid personality disorder include being suspicious that others are exploiting, harming, or deceiving them; reading hidden demeaning or threatening meanings into benign remarks/events; holding long-standing grudges; believing that others are not loyal or trustworthy; being reluctant to confide in others; feeling that others are attacking their character or reputation; having recurrent suspicions of infidelity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 560-562.

The correct answer is: Paranoid

Question 26

Not answered

Marked out of 1.00

Presence of lymphocytic pleiocytosis with many red cells in the CSF, along with hyperintense appearance of left temporal lobe in T2 weighted MRI imaging and lateralised periodic discharges in EEG in a semi-conscious patient suggests

Select one:

- Pneumococcal meningitis
- Creutzfeldt Jakob disease
- Temporal lobe malignancy
- Subdural haemorrhage
- Herpes encephalitis

Your answer is incorrect.

Brain abnormalities may be visible on CT but are much more consistently seen on MRI with gadolinium (MRI is most sensitive method of detecting early lesions and is imaging of choice); brain MRI typically shows hyperintensities involving the cortex and white matter in T2 and FLAIR images - areas of contrast enhancement can also be present. While MRI is the preferred modality of imaging, early in the clinical course of the disease, MRI results may be negative. A negative MRI does not rule out HSV encephalitis.

T2-weighted MRI reveals hyperintensity corresponding to edematous changes in the temporal lobes (see the first two images in the link http://emedicine.medscape.com/article/341142-overview) with a predilection for the medial temporal lobes.

Cerebrospinal fluid (CSF) examination is the gold standard for diagnosis. CSF polymerase chain reaction (PCR) for HSV has a sensitivity and specificity of >95%; false negatives can occur in the first 72 hours of onset. CSF pleocytosis (increased lymphocytes) and raised protein may also be seen.

Ref: Stahl JP, Mailles A. Herpes simplex virus encephalitis update. Current opinion in infectious diseases. 2019 Jun 1;32(3):239-43.

Kapadia RK et al. Encephalitis in adults caused by herpes simplex virus. CMAJ. 2020 Aug 10;192(32):E919-.

The correct answer is: Herpes encephalitis

Question 27

Not answered

Marked out of 1.00

Degeneration of head of caudate nucleus is associated with

Select one:

- Huntington chorea
- OCD
- Tourette syndrome
- Fahr disease
- Wilson disease

Your answer is incorrect.

Explanation:

In Huntington disease (HD), there is commonly mild-moderate cerebral atrophy. The main neuropathological abnormality of HD is atrophy of the caudate nucleus and putamen.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 206.

The correct answer is: Huntington chorea

Question 28

Not answered

Marked out of 1.00

Which one of the following is not a type of primary delusion?

Select one:

- Delusional memory
- Delusional misidentification
- Delusional atmosphere
- Delusional perception
- Autochthonous delusions

Your answer is incorrect.

Explanation:

Primary delusions are the direct result of psychopathology, while secondary delusions can be understood as having arisen in response to other primary psychiatric conditions. Primary delusions can be subdivided by the method by which they are perceived as having arisen, as follows:

1.Autochthonous delusion: arises 'out of the blue'

2.Delusional perception: arises on seeing a normal percept

3.Delusional memory: arises on recalling a memory

4.Delusional mood or atmosphere: arises on a background of anticipation, odd experiences, and increased awareness

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 106.

The correct answer is: Delusional misidentification

Question 29

Not answered

Marked out of 1.00

A 19 year-old university student, seen for first time by a psychiatrist, is very perplexed and claims that he could feel something odd happening around him. This phenomenon is termed as

Select one:

- Sensitive ideas of reference
- Delusional mood
- Oneiroid state
- Vorbeireden
- Persecutory delusion

Your answer is incorrect.

Explanation:

Delusional mood is recalled as arising following a period when there is an abnormal mood state characterised by anticipatory anxiety, a sense of 'something is about to happen' and an increased sense of the significance of minor events. The development of the formed delusion may come as a relief to the patient in this situation. Delusional mood is a type of Jasperian primary delusion.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

The correct answer is: Delusional mood

Question 30

Not answered

Marked out of 1.00

A teacher rewarded Jimmy with star sticker for his good behaviour. It represents which type of learning?

Select one:

- Operant conditioning
- Classical conditioning
- Social learning
- Aversive learning
- Cognitive learning

Your answer is incorrect.

Explanation:

Behaviour modification is based on operant conditioning. The basic principles of operant conditioning are that if a response to a stimulus produces positive consequences for the individual, it will tend to be repeated, while if it is followed by negative consequences, it will not. Forms of behavioural modification include the use of secondary reinforcement, such as 'token economy', in which socially desirable/acceptable behaviours are rewarded with tokens that can be exchanged for other material items or privileges, or 'star charts' where children's good behaviour is rewarded when a certain level is achieved. Rewarding a child with star stickers for good behaviour is an example of positive reinforcement.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 907-908.

The correct answer is: Operant conditioning

Question 31

Not answered

Marked out of 1.00

Mohammed is a 7-year-old boy. His mother told him that he could play outside once he had finished his homework. This is based on which of the following psychological mechanisms?

Select one:

- Premack principle
- Operant conditioning
- Reciprocal inhibition
- Classical conditioning
- Cognitive mapping

Your answer is incorrect.

Explanation:

The Premack principle asserts that access to a preferred action will reinforce any action that is less preferred. Any behaviour that is preferred to a second behaviour will theoretically reinforce the second behaviour. This is also known as ‘Grandma‘s rule' – ‘eat the carrot and get the dessert'.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1019.

The correct answer is: Premack principle

Question 32

Not answered

Marked out of 1.00

A grandmother says to her grandchild "You must eat your greens before you go to play outside". Which psychological mechanism is used here?

Select one:

- Premack principle
- Reciprocal inhibition
- Counterconditioning
- Overt sensitization
- Classical conditioning

Your answer is incorrect.

Explanation:

The Premack principle asserts that access to a preferred action will reinforce any action that is less preferred. Any behaviour that is preferred to a second behaviour will theoretically reinforce the second behaviour. The Premack principle is also called relative theory of reinforcement, often used with young children. It states that more probable behaviour (like play outside) will reinforce less probable behavior (like eating green leafy vegetables).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1019.

The correct answer is: Premack principle

Question 33

Not answered

Marked out of 1.00

Abnormalities in Stroop test suggests

Select one:

- Cortical blindness
- Lesion of optic chiasma
- Attentional bias
- Short term memory disturbance
- Damage to occipital cortex

Your answer is incorrect.

Explanation:

The Stroop effect is the delay and disruption in naming the colours of words printed in non-matching coloured ink, as when the word 'red' is printed in blue ink, the word 'blue' in green ink, and so on. To perform the task it is necessary to ignore the meanings of the printed words and to respond only to the colours in which they are printed, but with experienced adult readers automatic processing tends to occur, causing interference in the processing of information and a significant increase in the time required to name the colours, compared with the time taken to name the colours of meaningless strings of letters. Abnormal Stroop effect with the loss of appropriate inhibitory control is seen in frontal lobe lesions.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Attentional bias

Question 34

Not answered

Marked out of 1.00

One of the local newspaper reports that drug addicts carry out muggings. An 83-year-old lady was mugged and the public concluded the attacker was a drug addict. Select the cognitive heuristic in this case.

Select one:

- Framing
- Availability
- Simulation
- Representation
- Anchoring

Your answer is incorrect.

Explanation:

A heuristic is a rough-and-ready procedure or rule of thumb for making a decision, forming a judgement, or solving a problem without the application of an algorithm or an exhaustive comparison of all available options, and hence without any guarantee of obtaining a correct or optimal result. The availability heuristic is a cognitive heuristic through which the frequency or probability of an event is judged by the number of instances of it that can be readily brought to mind. It can generate biased or incorrect judgements.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015

The correct answer is: Availability

Question 35

Not answered

Marked out of 1.00

Which of the following is least likely to accompany lesions causing predominantly an impairment of working memory?

Select one:

- Procedural memory impairment
- Inability to learn new information
- Impaired performance on mathematical calculations
- Episodic memory impairment
- Impaired performance on tests of attention and concentration

Your answer is incorrect.

Explanation:

The performance of tasks such as typing, swimming, or cutting a loaf of bread are expressions of prior learning that differ from episodic memory in that there is no active awareness that memory is being searched in undertaking the particular skill. This type of memory is known as implicit, procedural, or skills memory. Studies of people with injury to the hippocampus suggest that declarative and implicit memory use different parts of the brain and can function independently. The hippocampus is believed to be the site where explicit (declarative) memory is stored, while implicit (procedural) memory is thought to reside in the limbic system, the amygdala, and the cerebellum.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 64.

The correct answer is: Procedural memory impairment

Question 36

Not answered

Marked out of 1.00

A patient with phobia for spiders is gradually exposed to various grades of anxiety provoking stimuli with simultaneous relaxation training. Which of the following principles are primarily employed in this therapy?

Select one:

- Discriminant learning
- Reciprocal inhibition
- Classical conditioning
- Covert desensitisation
- Stimulus generalisation

Your answer is incorrect.

Explanation:

Systematic desensitisation is based on the behavioural principle of **reciprocal inhibition** (i.e. anxiety and relaxation cannot coexist). It involves **three steps:** relaxation training, constructing a hierarchy of anxieties, and desensitisation of the stimulus. The patient is exposed to a **graded hierarchy of anxiety-provoking situations** in stepwise fashion while being taught to practice the relaxation techniques (e.g. a young woman with a spider phobia is taught relaxation techniques and then exposed to experiences starting from the lowest level of her hierarchy of anxieties - such as going to the cellar where there is the potential to come across a spider - then progressively going up the ladder to the highest - such as handling a spider).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 908.

The correct answer is: Reciprocal inhibition

Question 37

Not answered

Marked out of 1.00

During psychometric testing by a psychologist, the subject is first asked to copy a complex geometric figure and then to draw from memory after an interval of 30 minutes? What is this test called as?

Select one:

- Wechsler Adult Intelligence Scale
- Rivermead Behavioural Memory Test
- Wechsler Memory Scale
- Halstead-Reitan scale
- Rey-Osterrieth test

Your answer is incorrect.

Explanation:

The Rey-Osterrieth Complex Figure test examines the ability to draw and later recall a complex geometric configuration. It tests visual memory as well as executive deficits in the development of strategies and planning.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 53.

The correct answer is: Rey-Osterrieth test

Question 38

Not answered

Marked out of 1.00

Many patients with anxiety disorders engage in certain subtle behaviours while in a fearful situation that ultimately maintain the fear. These behaviours are called

Select one:

- Compulsions
- Safety behaviours
- Mental imagery
- Obsessions
- Phobic behaviours

Your answer is incorrect.

Explanation:

Safety behaviours are overt or covert acts intended to prevent a feared outcome or to minimise its consequences. They are found in many anxiety disorders, including PTSD, OCD, panic disorder, and social anxiety disorder (SAD). Safety behaviours such as avoiding eye contact or rehearsing sentences before saying them are used in order to prevent feared outcomes such as public embarrassment or humiliation. However, as the fears are excessive, it has been suggested that a major effect of safety behaviours is that they prevent patients from disconfirming some of their erroneous beliefs about the feared situation.

Gray E et al. Sub-types of safety behaviours and their effects on social anxiety disorder. Plos one. 2019 Oct 1;14(10):e0223165.

The correct answer is: Safety behaviours

Question 39

Not answered

Marked out of 1.00

During cognitive examination, a patient correctly states that bicycles have two wheels. What type of memory has been tested?

Select one:

- Semantic memory
- Episodic memory
- Procedural memory
- Working memory
- Autobiographical memory

Your answer is incorrect.

Explanation:

Semantic memory is a type of long-term memory for factual information about the world, excluding personal episodes in one's life. Typical examples are knowledge of the dates of the world wars and the chemical formula for water. The term is from the Greek 'semantikos', so called because it is necessary for the use of language. Episodic or autobiographic memory involves recalling an 'episode' - the time associated memory.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Semantic memory

Question 40

Not answered

Marked out of 1.00

Sammy Taylor is coming to see you in clinic as an emergency book-in. He is 19 years old and has developed some voice-hearing experiences and bizarre thoughts regarding the MI5. Which of these might be a precipitating factor?

Select one:

- Having lost his adoptive father to a road traffic accident two weeks previously
- Having been adopted at the age of 7 months
- Having had a traumatic birth
- Having a brother who has a diagnosis of schizophrenia
- Having dropped out of school prematurely at the age of 15

Your answer is incorrect.

Explanation:

The others may be contributing factors, but most are predisposing rather than precipitating. A precipitating factor has a strong temporal link to the onset of the episode of mental illness (e.g. it occurs just before).

Four 'Ps' of case formulation:

Predisposing factors: those that put a person at risk of developing a problem; may include genetics, previous life events, or temperament

Precipitating factors: a specific event or trigger to the onset of the current problem

Perpetuating factors: those that maintain the problem once it has become established

Protective factors: strengths that reduce the severity of problems and promote healthy and adaptive functioning

Ref: Racine NM et al. Systematic review: predisposing, precipitating, perpetuating, and present factors predicting anticipatory distress to painful medical procedures in children. Journal of pediatric psychology. 2016 Mar 1;41(2):159-81.

The correct answer is: Having lost his adoptive father to a road traffic accident two weeks previously

Question 41

Not answered

Marked out of 1.00

The current international agreement on the research of human subjects is

Select one:

- Tuskegee syphilis study
- Geneva declaration
- Hippocratic oath
- Nuremberg code
- Helsinki declaration

Your answer is incorrect.

Explanation:

The World Medical Association (WMA) has developed the Declaration of Helsinki as a statement of ethical principles for medical research involving human subjects, including research on identifiable human material and data. Adopted by the WMA's General Assembly in 1964, it is the most important set of guidelines about research on human participants. It both reflects and shapes the ethos of international research ethics. It is a living instrument and is reviewed and revised regularly. Its latest revision was in 2013.

Ref: https://www.wma.net/policies-post/wma-declaration-of-helsinki-ethical-principles-for-medical-research-involving-human-subjects/

Malik AY, Foster C. The revised Declaration of Helsinki: cosmetic or real change?. Journal of the Royal Society of Medicine. 2016 May;109(5):184-9.

The correct answer is: Helsinki declaration

Question 42

Not answered

Marked out of 1.00

An Inuit woman suffers from sudden attacks of shouting, becoming frightened and impulsive running away. She appears dissociated during the attacks and has total amnesia for the episodes. Probable diagnosis is

Select one:

- Windigo
- Dhat
- Latah
- Fag
- Piblokto

Your answer is incorrect.

Explanation:

Piblokto is a culture-specific syndrome found among the Inuit in which individuals present with attacks of screaming, crying, and running naked through the snow. It is characterised by an initial period of fatigue, social withdrawal, confusion, and irritability lasting for a few hours or days, leading up to an abrupt episode of dissociation (that usually lasts only minutes) during which the affected person may strip or tear off clothes, roll in the snow, run about in a frenzied state, shout obscenities, destroy property, engage in other violent, dangerous, or antisocial forms of behaviour, and manifest pseudolalia, echolalia, and echopraxia, before lapsing into convulsions and losing consciousness. This is followed by complete amnesia for the dissociative episode.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 126.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Piblokto

Question 43

Not answered

Marked out of 1.00

A 17-year-old man is using cannabis for last 6 months. His parents worry about his dependence but he is not worried. Which stage of Prochaska & DiClemente's transtheoretical model of change is he presently in?

Select one:

- Pre contemplation
- Maintenance
- Contemplation
- Action
- Preparation

Your answer is incorrect.

Explanation:

The most appropriate stage of change in the transtheoretical model is precontemplation stage, as he is not even ready to think about change seriously.

Six stages of change:

(1) **Precontemplation:** not even considering changing behaviour; does not see the behaviour as a problem, minimizes and denies associated risks, and avoids information to the contrary

(2) **Contemplation:** has become aware that the behaviour is a problem but is ambivalent about changing, and likely sees equal or more benefits than costs from the behaviour

(3) **Preparation:** has made a decision to attempt change

(4) **Action:** has implemented a plan and is changing the behaviour

(5) **Maintenance:** maintaining gains made and attempting to improve areas of life that have been harmed

(6) **Relapse:** return to previous behaviour, but with the possibility of gaining useful strategies to extend the maintenance period on the next attempt

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 575.

The correct answer is: Pre contemplation

Question 44

Not answered

Marked out of 1.00

What ethical document, sometimes referred to as the 'Physician's Pledge' or the 'Modern Hippocratic Oath', was adopted by the World Medical Association in 1948?

Select one:

- The Declaration of Malta
- The Declaration of Tokyo
- The Declaration of Taipei
- The Declaration of Geneva
- The Declaration of Helsinki

Your answer is incorrect.

Explanation:

The Declaration of Geneva represents the important principles of professional medical conduct in the modern globalised world. It was last amended in 2017. It is written in the tradition of the Hippocratic Oath, but unlike the Hippocratic Oath it contains no reference to God, no prohibition of surgery, no explicit restrictions on euthanasia or abortion, but simply a commitment to the 'utmost respect for human life'. The Hippocratic Oath prohibits abuse, whilst the Declaration of Geneva prohibits discrimination. The 2017 version of the Geneva Declaration mentions patient autonomy for the first time. The Helsinki declaration refers to a statement of ethical principles for medical research involving human subjects. Other options are not relevant for medical ethics.

Ref: Wiesing U. The Hippocratic Oath and the Declaration of Geneva: legitimisation attempts of professional conduct. Medicine, Health Care and Philosophy. 2020 Mar;23(1):81-6.

The correct answer is: The Declaration of Geneva

Question 45

Not answered

Marked out of 1.00

Which of the following statements is true with regard to public and professional perception of depressive illness?

Select one:

- Professionals do not perceive antidepressants to be very helpful for depression
- Health professionals and public hold concordant views on management of depression
- Public have greater faith in antidepressants than the professionals
- Public does not support special diet or nutrition to treat depression
- Public perceive psychotropics to be more harmful than helpful

Your answer is incorrect.

Health professionals give higher ratings than the public to the helpfulness of antidepressants for depression, and of antipsychotics and admission to a psychiatric ward for schizophrenia. Conversely, the public tended to give much more favourable ratings to vitamins and minerals and special diets for both depression and schizophrenia, and to reading self-help books for schizophrenia. The beliefs that health practitioners hold about mental disorders differ greatly from those of the general public. (Retrieved from Jorm AF, Korten AE, et al. Helpfulness of interventions for mental disorders: beliefs of health professionals compared with the general public. Br J Psychiatry 1997; 171: 233-237)

The correct answer is: Public perceive psychotropics to be more harmful than helpful

Question 46

Not answered

Marked out of 1.00

Which of the following is a component of Kohlberg's conventional stage of morality?

Select one:

- Authority
- Punishment
- Reward
- Social contract
- Universal ethics

Your answer is incorrect.

Explanation:

Lawrence Kohlberg integrated Jean Piaget's concepts and described multiple stages of moral development within three significant levels of morality. The first level is pre-conventional morality, in which punishment and obedience are the determining factors. The second level is morality of conventional role-conformity, in which children try to conform to gain approval and maintain good relationships with others and are also guided by rigid codes of 'law and order' (authority orientation). The third and highest level is the morality of self-accepted moral principles (post-conventional), in which children voluntarily comply with rules based on a concept of universal ethical principles and make exceptions to rules in certain circumstances.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 902.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Authority

Question 47

Not answered

Marked out of 1.00

According to Erikson's stages of psychosocial development, at which stage would a four year old be?

Select one:

- Initiative vs guilt
- Industry vs inferiority
- Trust vs mistrust
- Autonomy vs shame
- Ego identity vs role confusion

Your answer is incorrect.

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. Stage 3 takes place from about 3 years to about 5 years. Its focus is initiative versus guilt. The child's increasing mastery of locomotor and language skills expands its participation in the outside world and stimulates omnipotent fantasies of broader exploration and conquest. Guilt over the drive for conquest and anxiety over the anticipated punishment are both assuaged in the child through repression of the forbidden wishes and development of a superego to regulate its initiative. Success in this stage will result in development of the basic virtue of purpose.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1073-1074.

The correct answer is: Initiative vs guilt

Question 48

Not answered

Marked out of 1.00

Mrs. Kean and her 16-year-old son Henry have been having some disagreements about his future plans. Mrs. Kean has decided which universities Henry will apply for. He protests, saying he wants to make his own decision, but Mrs. Kean replies saying that the choice is not his, the decision will be made for him by her and her husband and there's nothing he can do about it. Henry reluctantly concedes. What parenting style is Mrs. Kean exhibiting?

Select one:

- Authoritarian
- Boundaried
- Authoritative
- Neglectful
- Permissive

Your answer is incorrect.

Explanation:

Rutter has clustered parenting into four general styles. The authoritarian style, characterised by strict, inflexible rules, can lead to low self-esteem, unhappiness, and social withdrawal. Authoritarian parents tend to be restrictive and cold, and their children tend to be withdrawn or conflicted.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 898.

The correct answer is: Authoritarian

Question 49

Not answered

Marked out of 1.00

Melissa and her partner Jakob have recently had a baby. Jakob is looking forward to charting their son's development, and wants to draw up a time-line so that he can anticipate what new skills his son may be able to master. He is anxious to notice any clues that may suggest his son is not developing as expected. Which of the following scenarios suggests impaired development?

Select one:

- 2-3 words at 24 months
- Holding head up at 3 months
- Able to roll at 9 months
- Able to walk at 13 months
- Plays with a ball at 16 months

Your answer is incorrect.

Explanation:

By age 18 months, a child is generally using 20 single words. They will understand up to 150 words. By 24 months, their vocabulary will increase to up to 270 words.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 889.

The correct answer is: 2-3 words at 24 months

Question 50

Not answered

Marked out of 1.00

What type of attachment is demonstrated by a child who seems unaffected by the mother leaving and carries on playing and also unaffected by the mother's return?

Select one:

- Secure attachment
- Anxious-avoidant attachment
- Anxious-ambivalent attachment
- Anxious-resistant attachment
- Disorganised attachment

Your answer is incorrect.

Explanation:

Mary Ainsworth devised the Strange Situation procedure with 12- to 18-month infants, which categorised infant-parent relationships. The procedure involves separations and reunions with caregivers and observes response. In anxious-avoidant attachment, the child seems unconcerned at separation or reunion. It is associated with unresponsive parenting. Mary Ainsworth noted that this type of attachment often occurred when the child's needs were repeatedly not met and allows the child to maintain some proximity to the parent without experiencing maternal rejection. Note that the anxious-avoidant, anxious-resistant and disorganised categories are also described as insecure-avoidant, insecure-resistant (or ambivalent) and insecure-disorganised. Ainsworth preferred classifications using alphabetical notations (A, B, C). The disorganised category is attributed to Ainsworth's student, Mary Main.

For a short illustrative video see

Ref: Duschinsky R. Mary Ainsworth and the Strange Situation Procedure. In: *Cornerstones of Attachment Research*. New York, NY: Oxford University Press; 2020:109-210. https://www.oxfordclinicalpsych.com/view/10.1093/med-psych/9780198842064.001.0001/med-9780198842064-chapter-2. Accessed March 4, 2022.

The correct answer is: Anxious-avoidant attachment

Question 51

Not answered

Marked out of 1.00

Nina and her mother are spending the afternoon together. Nina's mother has been talking about families. At one point in the conversation Nina asks her mother 'what would happen if I have a baby brother?' What developmental stage, according to Piaget, is Nina most likely to be at?

Select one:

- Formal Operational
- Preoperational
- Concrete Operational
- Conventional
- Sensorimotor

Your answer is incorrect.

Explanation:

According to Piaget, the stage of formal operations lasts from about 11 years of age through the end of adolescence. The stage is so named because young persons' thinking operates in a formal, highly logical, systematic, and symbolic manner. It is characterised by the ability to think abstractly, to reason deductively, and to define concepts, and also by the emergence of skills for dealing with permutations and combinations. Adolescents attempt to deal with all possible relations and hypotheses to explain data and events during this stage. They are able to think abstractly or hypothetically, which is demonstrated by Nina's question.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1009.

The correct answer is: Formal Operational

Question 52

Not answered

Marked out of 1.00

Kevin is a young boy who is getting into arguments with other children at school and is occasionally pushing them over. His parents and teachers have attempted to reason with him but this has not been successful. His behaviour eventually improves when he is told that his favourite toy will be removed as long as he continues to push others. Which of the following stages of Kohlberg's moral development would best fit this scenario?

Select one:

- Obedience and punishment orientation
- Authority and social-order maintaining orientation
- Self-interest orientation
- Social contract orientation
- Interpersonal accord and conformity

Your answer is incorrect.

Explanation:

Lawrence Kohlberg integrated Jean Piaget's concepts and described multiple levels of moral development within three significant levels of morality. The first level is pre-conventional morality, in which punishment and obedience are the determining factors. The second level is morality of conventional role-conformity, in which children try to conform to gain approval and maintain good relationships with others and are also guided by rigid codes of 'law and order' (authority orientation). The third and highest level is the morality of self-accepted moral principles (post-conventional), in which children voluntarily comply with rules based on a concept of universal ethical principles and make exceptions to rules in certain circumstances.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 902.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Obedience and punishment orientation

Question 53

Not answered

Marked out of 1.00

In the development of vision, which of the following can be seen occurring as early as the first day after birth?

Select one:

- Pupil reaction to light
- Accommodation
- Convergence
- Colour vision
- Depth perception

Your answer is incorrect.

The pupillary light reflex (PLR) whereby the diameter of the pupil is controlled according to the intensity of light occurs from birth. Depth perception, colour vision and accommodation occur at three to five months. Convergence occurs at five to seven months.

The correct answer is: Pupil reaction to light

Question 54

Not answered

Marked out of 1.00

Mrs Molloy has been struggling with feelings of depression and self-blame. She's been talking about her childhood experiences in psychotherapy and described the way her father would make her follow very strict rules. On one occasion, she returned home only a few minutes after her curfew, and she was beaten by him with a belt. The parenting style she is describing is known as

Select one:

- Authoritarian
- Permissive
- Indulgent
- Neglectful
- Secure

Your answer is incorrect.

Explanation:

Rutter has clustered parenting into four general styles. The authoritarian style, characterised by strict, inflexible rules, can lead to low self-esteem, unhappiness, and social withdrawal. Authoritarian parents tend to be restrictive and cold, and their children tend to be withdrawn or conflicted. These children generally tend to grow up with little sense of agency or personal power, with a tendency towards depression, self-doubt and fear.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 898.

The correct answer is: Authoritarian

Question 55

Not answered

Marked out of 1.00

You are about to go to A&E to assess a patient who is feeling suicidal. Which demographic description has got the strongest association with completed suicide?

Select one:

- Middle-aged male
- Elderly female
- Young female
- Widow
- Elderly male

Your answer is incorrect.

Explanation:

In the UK, men aged 45-49 years currently present the highest suicide rate. This is thought to be linked to middle-aged men being more likely affected in recent years by economic adversity, alcoholism, and social isolation. In most countries, rates of suicide are 2-3 times higher in males than in females, which could possibly be due to a male preference for higher lethality methods and the reluctance of males to seek help.

Ref: Correia R, Jackson D. Risk to self: identifying and managing risk of suicide and self-harm. BJPsych Advances. 2020:1-11.

Turecki G et al. Suicide and suicide risk. Nature reviews Disease primers. 2019 Oct 24;5(1):1-22.

The correct answer is: Middle-aged male

Question 56

Not answered

Marked out of 1.00

The mode of language designed to provide a means of communication for those with learning difficulties is

Select one:

- COPE method
- BASDEC method
- Jarmen system
- Makaton system
- Barthel method

Your answer is incorrect.

Makaton is a simplified sign language, based on British Sign Language but internationally recognised, specifically designed for people with learning difficulties of all kinds and their parents and carers, developed by the speech therapist Margaret Walker. The revised version contains approximately 350 signs and associated words. 'Makaton' is formed from the initial letters of the first names of its inventor, Margaret Walker, and the two psychiatric hospital visitors who assisted in its development, Kathy Johnston and Tony Cornforth.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Makaton system

Question 57

Not answered

Marked out of 1.00

Who proposed a theory of identity development in children, separate from that of their mothers' identity?

Select one:

- Freud
- Rutter
- Margaret Mahler
- Spitz
- Winnicott

Your answer is incorrect.

Explanation:

Margaret Mahler proposed a theory to describe how young children acquire a sense of identity separate from that of their mothers'. Her observations of children and their mothers lead to her theory of separation-individuation.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 892.

The correct answer is: Margaret Mahler

Question 58

Not answered

Marked out of 1.00

Which of the following type of parents will not set any limit for child's behaviour but show unpredictable harshness and act aggressively if boundaries are violated?

Select one:

- Authoritarian
- Authoritative
- Permissive
- Good enough parent
- Neglectful

Your answer is incorrect.

Explanation:

Rutter has clustered parenting into four general styles. The indulgent-permissive style, which includes little or no limit-setting coupled with unpredictable parental harshness, can lead to low self-reliance, poor impulse control, and aggression.

Authoritarian style parents impose strict rules that can be associated with low self-esteem and social withdrawal in the children. The neglectful style of parenting can lead to poor self-esteem, impaired self-control, and increased aggression. Parents with an authoritative style set firm rules but share decision-making with their children, creating a warm, loving environment, which promotes self-reliance and positive self-esteem.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 898.

The correct answer is: Permissive

Question 59

Not answered

Marked out of 1.00

A 4-year-old child now recognizes she's a girl and wants to spend more time with her dad. According to Freud she's at what phase of psychosexual development?

Select one:

- Genital phase
- Oral phase
- Anal phase
- Latency phase
- Phallic phase

Your answer is incorrect.

Explanation:

The phallic stage begins sometime during year 3 and continues until approximately the end of year 5. The phallic stage provides the foundation for an emerging sense of sexual identity, curiosity without embarrassment, initiative without guilt, as well as mastery not only over objects and persons in the environment but also over internal processes and impulses.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1062-1063.

The correct answer is: Phallic phase

Question 60

Not answered

Marked out of 1.00

Angela Stubbs has been participating in psychotherapy for the past seven months. She has the sense that the process has been helping her develop a more robust sense of self, a feeling of continuity, and of affiliation with others. What term best describes this process?

Select one:

- Individuation
- Identity diffusion
- Symbiosis
- Introjection
- Deindividuation

Your answer is incorrect.

Explanation:

Carl Jung described individuation as “becoming a single, homogeneous being, and, in so far as ‘individuality’ embraces our innermost, last, and incomparable uniqueness, it also implies becoming one’s own self. We could therefore translate individuation as ‘coming to selfhood’ or ‘self-actualisation’."

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Individuation

Question 61

Not answered

Marked out of 1.00

Mr. Ellison is a 70-year-old man who was taken to his GP with increasing symptoms of cognitive impairment, resting tremor, limb stiffness and episodes when he appears to be seeing distressing things. Which of the following is the gene product most likely to be associated with his diagnosis?

Select one:

- Alpha-synuclein
- Ubiquitin
- Beta-Synuclein
- Tau protein
- Amyloid A protein

Your answer is incorrect.

Explanation:

The diagnosis is Lewy body dementia (LBD) and the pathological finding in this condition is the presence of Lewy bodies. The key constituent of Lewy bodies is alpha-synuclein. The function of alpha-synuclein is not certain, but it is thought to function in cell membrane remodeling at neuronal terminals. When aggregates of misfolded, overexpressed alpha-synuclein accumulate in neurons or surrounding glial cells, Lewy bodies are formed.

Ref: Sanford AM. Lewy body dementia. Clinics in geriatric medicine. 2018 Nov 1;34(4):603-15.

The correct answer is: Alpha-synuclein

Question 62

Not answered

Marked out of 1.00

Abdul is a 10-year-old biy who suffers from ADHD. You meet with his parents in your outpatient clinic. They have been reading about Modafinil on the Internet and are keen to learn more. Through which neurotransmitter does modafinil predominantly exert its effect in ADHD?

Select one:

- Dopamine
- GABA
- Serotonin
- Glutamate
- Acetylcholine

Your answer is incorrect.

Explanation:

How modafinil works is unknown, but it is clearly different from classical stimulants such as methylphenidate and amphetamine. Modafinil binds to and requires the presence of the dopamine transporter; it also requires the presence of alpha-adrenergic receptors. It hypothetically acts as an inhibitor of the dopamine transporter.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 521.

The correct answer is: Dopamine

Question 63

Not answered

Marked out of 1.00

The movement of ions is an essential part of the transmission of impulses within the nervous system. Which of the following ions flows into neurons to propagate action potentials?

Select one:

- Sodium
- Potassium
- Chloride
- Phosphate
- Calcium

Your answer is incorrect.

Explanation:

An action potential is the momentary change in the voltage difference across the membrane of a neuron, usually triggered by nerve impulses from several other neurons at synapses. It is characterised by an influx of positively charged sodium ions through the cell membrane, resulting in an impulse being propagated along the neuron's axon, reversing its resting potential of approximately -70 millivolts (mV), the inner surface being negative relative to the outer surface and the membrane more permeable to potassium ions than to sodium ions, and changing it to approximately +40 mV, increasing the permeability of the membrane to sodium ions and resulting in the influx of positively charged ions, thereby propagating the potential from the axon hillock down the axon.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Sodium

Question 64

Not answered

Marked out of 1.00

Alzheimer disease is strongly associated with:

Select one:

- Presenilin-1 gene on chromosome 1
- Presenilin-2 gene on chromosome 2
- COMT gene on chromosome 21
- CAG repeats on chromosome 4
- APO E4 on chromosome 19

Your answer is incorrect.

Explanation:

Carriers of the apolipoprotein E E4 allele on chromosome 19 are at increased risk of Alzheimer disease (AD). Early onset AD has been associated with mutations including amyloid precursor protein on chromosome 21, and the genes for presenilin 1 (on chromosome 14) and presenilin 2 (on chromosome 1).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 156.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 993.

The correct answer is: APO E4 on chromosome 19

Question 65

Not answered

Marked out of 1.00

Which of the following is caused by trinucleotide repeat expansion?

Select one:

- Down syndrome
- Huntington disease
- Parkinson disease
- Schizophrenia
- Lewy body dementia

Your answer is incorrect.

Explanation:

Huntington disease results from expansion of a **CAG trinucleotide repeat** in the HTT (huntingtin) gene on the short arm of chromosome 4. Alleles with 40 or more trinucleotide repeats are fully penetrant.

Ref: Bachoud-Lévi AC et al. International Guidelines for the treatment of Huntington's Disease. Frontiers in neurology. 2019;10:710.

The correct answer is: Huntington disease

Question 66

Not answered

Marked out of 1.00

A man is brought to the A& E department with akinetic mutism. He has suffered a stroke. Which artery is most likely to be involved?

Select one:

- Basilar artery
- Subclavian artery
- Posterior inferior cerebellar artery
- Middle cerebral artery
- Bilateral anterior cerebral artery

Your answer is incorrect.

Explanation:

Damage to the anterior cingulate cortex is known to be a primary cause for akinesia and mutism and has frequently been described in case reports of patients with bleeding, stroke, or other types of severe brain injury. The most dramatic examples of akinesia and mutism follow bilateral lesions of the anterior cingulate cortex, such as after bilateral damage to the vascular supply of the anterior cerebral artery territory.

Ref: Arnts H et al. On the pathophysiology and treatment of akinetic mutism. Neuroscience & Biobehavioral Reviews. 2020 May 1;112:270-8.

The correct answer is: Bilateral anterior cerebral artery

Question 67

Not answered

Marked out of 1.00

Communication between the frontal lobes of each side of the brain is mediated through which of the following structures?

Select one:

- Fornix
- Genu of corpus callosum
- Parahippocampal gyrus
- Splenium of corpus callosum
- Arcuate fasciculus

Your answer is incorrect.

The corpus callosum forms the major connection by which the hemispheres communicate with one another. The genu of the corpus callosum contains fibers that pass from the frontal lobe of one side to that of the other.

The correct answer is: Genu of corpus callosum

Question 68

Not answered

Marked out of 1.00

Which of the following structures appears larger on the left side than the right in healthy right-handed man?

Select one:

- Frontal lobe
- Hippocampus
- Inferior temporal gyrus
- Caudate nucleus
- Heschl's gyrus

Your answer is incorrect.

Explanation:

Heschl’s gyrus (HG), or the first transverse temporal gyrus, first described by Richard Heschl, is an oblique convolution located on the inferior surface of the lateral fissure (also known as the Sylvian fissure), which runs transversely (mediolaterally) from near the insula towards the lateral part of the superior temporal gyrus. HG includes the primary auditory cortex, this being the first cortical relay station of auditory information in the brain. On average and in healthy populations, HG is large in the left compared to the right hemisphere.

Ref: da Rocha JL et al. TASH: Toolbox for the Automated Segmentation of Heschl’s gyrus. Scientific reports. 2020 Mar 3;10(1):1-5.

The correct answer is: Heschl's gyrus

Question 69

Not answered

Marked out of 1.00

A 40-year-old man develops depressed mood, anhedonia, initial and terminal insomnia, loss of appetite, a 10-lb weight loss, and difficulty with sexual arousal. The clinical features of the patient's psychiatric illness suggest dysfunction of the

Select one:

- Hippocampus
- Corpus callosum
- Frontal lobes
- Pituitary
- Hypothalamus

Your answer is incorrect.

Clinical studies of patients with major depressive disorders indicate that an intrinsic regulatory defect involving the hypothalamus underlies the disorder. It also involves the monoamine pathways. The hypothalamic modulation of neuroendocrine activity has been implicated, as have been the neurotransmitter systems of serotonin and norepinephrine.

The correct answer is: Hypothalamus

Question 70

Not answered

Marked out of 1.00

Which of the following statements about microglia is correct?

Select one:

- It is mainly responsible for myelination
- It constitutes 70% of glial population in brain
- It is sensitive to sodium channel stimulation
- It is mesodermal in origin
- It is present outside brain and spinal cord

Your answer is incorrect.

Explanation:

Microglia, which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance. Because of their mesodermal origin, microglia share many features with other myeloid cell types in the body.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

Ginhoux F, Prinz M. Origin of microglia: current concepts and past controversies. Cold Spring Harbor perspectives in biology. 2015 Aug 1;7(8):a020537.

The correct answer is: It is mesodermal in origin

Question 71

Not answered

Marked out of 1.00

Which of the following is the primary mechanism involved in removal of the transmitter at the neuromuscular junction?

Select one:

- Reuptake
- Enzymatic degradation
- Diffusion
- Immune complex formation
- Renal clearance

Your answer is incorrect.

Explanation:

A transmitter can be removed from the synaptic cleft through one of the following three modes: (1) enzymatic degradation, (2) reuptake, and (3) diffusion. In neuromuscular junctions acetylcholine, the primary neurotransmitter, undergoes enzymatic degradation. Acetylcholinesterase is an essential enzymatic component of the neuromuscular junction where it is responsible for terminating neurotransmission by the cholinergic motor neurons. The enzyme at the neuromuscular junction is contributed primarily by the skeletal muscle where is it produced at higher levels in the post-synaptic region of the fibres.

Ref: Rotundo RL. The NMJ as a model synapse: New perspectives on formation, synaptic transmission and maintenance: Acetylcholinesterase at the neuromuscular junction. Neurosci Lett. 2020 Sep 14;735:135157. doi: 10.1016/j.neulet.2020.135157. Epub 2020 Jun 12. PMID: 32540360.

The correct answer is: Enzymatic degradation

Question 72

Not answered

Marked out of 1.00

Removal of noradrenaline from the region of the synaptic cleft may be achieved by which of the following mechanisms?

Select one:

- reuptake only
- diffusion only
- conversion to adrenaline
- enzymatic degradation only
- enzymatic degradation, diffusion, and reuptake

Your answer is incorrect.

Explanation:

Norepinephrine (NE) action is terminated by two principal destructive or catabolic enzymes that turn NE into inactive metabolites. The first is monoamine oxidase A or B, which is located in mitochondria in the presynaptic neuron and elsewhere. The second is catechol-O-methyltransferase (COMT), which is thought to be located largely outside of the presynaptic nerve terminal. The action of NE can be terminated not only by enzymes that destroy NE, but also by a transport pump for NE that removes it from acting in the synapse without destroying it. That transport pump is sometimes called the 'NE transporter' or 'NET' and sometimes the 'NE reuptake pump'.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 253-254.

The correct answer is: enzymatic degradation, diffusion, and reuptake

Question 73

Not answered

Marked out of 1.00

The metabotropic receptors are coupled to

Select one:

- Ligand non-dependent regulators of nuclear transcription
- G proteins
- Ligand dependent regulators of nuclear transcription
- Proteins that have intrinsic enzyme activity
- Ligand gated channels

Your answer is incorrect.

Explanation:

Metabotropic receptors are linked to G proteins.

Receptors are categorized into: (1) ionotropic: ligand-gated channels, in which binding of a chemical messenger alters the probability of opening of transmembrane pores or channels; (2) metabotropic: those in which the receptor proteins are coupled to intracellular G proteins as transducing elements (3) intrinsic enzymes: those consisting of single membrane-spanning protein units that have intrinsic enzyme activity (for example, having tyrosine kinase activity; some authors classify them alongside metabotropic receptors); and (4) ligand-dependent regulators of nuclear transcription (including receptors for corticosteroids such as testosterone)

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 100.

The correct answer is: G proteins

Question 74

Not answered

Marked out of 1.00

The neurotransmitter involved in the long-term potentiation of memory is

Select one:

- Noradrenaline
- Dopamine
- Acetyl choline
- Glutamate
- Serotonin

Your answer is incorrect.

Explanation:

Normally, when glutamate synapses are active, their NMDA receptors trigger an electrical phenomenon known as long-term potentiation (LTP). With the help of gene products that converge upon glutamate synapses and receptors, ion channels, and the processes of neuroplasticity and synaptogenesis, LTP normally leads to structural and functional changes of the synapse that make neurotransmission more efficient, sometimes called 'strengthening' of synapses.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 151.

The correct answer is: Glutamate

Question 75

Not answered

Marked out of 1.00

The neurotransmitter which is most widely distributed in the brain is

Select one:

- Dopamine
- Glutamate
- Acetylcholine
- Norepinephrine
- Serotonin

Your answer is incorrect.

Explanation:

The most prevalent neurotransmitter in the brain is glutamate. Glutamate is a ubiquitous excitatory neurotransmitter that seems to be able to excite nearly any neuron in the brain. That is why it is sometimes called the 'master switch'.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 102.

The correct answer is: Glutamate

Question 76

Not answered

Marked out of 1.00

In a patient being evaluated for neuropsychiatric disturbances, CT scan shows atrophy of head of caudate nucleus. Which of the following is a likely diagnosis?

Select one:

- Alzheimer disease
- Wilson disease
- Fahr disease
- Huntington disease
- Pick disease

Your answer is incorrect.

Explanation:

In Huntington disease (HD), there is commonly mild-moderate cerebral atrophy. The main neuropathological abnormality of HD is atrophy of the caudate nucleus and putamen.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 206.

The correct answer is: Huntington disease

Question 77

Not answered

Marked out of 1.00

Which of the following neuroimaging finding is most suggestive of new variant CJD?

Select one:

- Occipital hypoperfusion on HMPAO- SPECT
- Activation of primary auditory cortex on functional MRI
- Impaired dorsolateral cortex function
- Medial temporal atrophy on structural MRI scan
- Increased signal in the pulvinar nucleus of thalamus bilaterally

Your answer is incorrect.

Explanation:

In variant CJD (vCJD), cranial magnetic resonance imaging (MRI) scans show a symmetrical area of hyperintensity on FLAIR sequences in the posterior thalamus (the 'pulvinar sign'), which is highly characteristic and has been incorporated into the clinical diagnostic criteria for vCJD.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 169.

The correct answer is: Increased signal in the pulvinar nucleus of thalamus bilaterally

Question 78

Not answered

Marked out of 1.00

A patient develops viral encephalitis. He exhibits increased oral exploratory behaviour and inappropriate sexuality. This syndrome can be called

Select one:

- Kluver Bucy syndrome
- Korsakoff syndrome
- Lennox Gastaut syndrome
- Gerstmann syndrome
- Klein Levine syndrome

Your answer is incorrect.

Explanation:

Kluver-Bucy syndrome is an experimental model of temporal lobe ablation in monkeys. Behaviour in this syndrome is characterised by hypersexuality, placidity, a tendency to explore the environment with the mouth, inability to recognise the emotional significance of visual stimuli, and continually shifting attention ('hypermetamorphosis'). This resembles the presentation of people with bilateral injury to the temporal lobes after head trauma, cardiac arrest, herpes simplex encephalitis, or Pick disease (frontotemporal dementia).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 933.

The correct answer is: Kluver Bucy syndrome

Question 79

Not answered

Marked out of 1.00

A 44 year old man is treated for long standing asthma with a combination of oral theophylline, prednisolone, inhaled salbutamol and inhaled budesonide. He appears heavy with purplish striae on abdomen and excessive fat on the nape of his neck. His asthma is under control but he exhibits behaviours and mental state consistent with hypomania. Which of the following is a possible cause?

Select one:

- Late onset bipolar disorder
- Theophylline induced epileptiform activity
- Hypothyroidism
- Cushing syndrome
- Secondary diabetes mellitus

Your answer is incorrect.

Explanation:

The story is consistent with Cushing syndrome. Cushing syndrome most commonly results from exogenous corticosteroids. Physical symptoms may include diabetes, hypertension, muscle weakness, obesity, and osteopenia. Psychiatric symptoms may appear before physical signs; depression is most common (50-70% of cases), followed by anxiety, hypomania/mania, psychosis, and cognitive dysfunction. Exogenous steroids produce more mania than endogenous steroids, often dose-related. This case is most likely due to oral prednisolone, not inhaled budesonide.

Ref: Tang A et al. Psychiatric symptoms as a clinical presentation of Cushing's syndrome. Annals of general psychiatry. 2013 Dec;12(1):1-3.

The correct answer is: Cushing syndrome

Question 80

Not answered

Marked out of 1.00

During EEG recording a resting subject is asked to focus his attention on a cognitive task. Which of the following will be observed?

Select one:

- No change in resting EEG
- Unilateral change in resting EEG at dominant hemisphere
- Appearance of alpha waves on focussing attention
- Reduction of alpha waves on focussing attention
- Reduction of theta waves on focussing attention

Your answer is incorrect.

Explanation:

Alpha blocking, a phenomenon where the alpha rhythm is reduced by attention to a visual, auditory, tactile, or cognitive stimulus, is one of the most prominent features of human EEG signals.

Ref: Hartoyo A et al. Inferring a simple mechanism for alpha-blocking by fitting a neural population model to EEG spectra. PLoS computational biology. 2020 Apr 30;16(4):e1007662.

The correct answer is: Reduction of alpha waves on focussing attention

Question 81

Not answered

Marked out of 1.00

In the dementia associated with Alzheimer's disease, the EEG will usually show which of the following?

Select one:

- Periodic frontal lobe discharges
- Generalized background slowing
- Focal slowing
- An isoelectric record
- Spike-and-wave discharges

Your answer is incorrect.

Explanation:

Diffuse slowing of the background rhythm of the EEG is the most common EEG abnormality. Diffuse slowing in the EEG can be seen in dementia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 999.

The correct answer is: Generalized background slowing

Question 82

Not answered

Marked out of 1.00

Bill Bailey is a 42-year-old man with a diagnosis of schizophrenia, whose functioning has been severely compromised for the past decade, despite long trials of numerous medications. Three months ago he was commenced on Clozapine, but it hasn't had the ameliorating effects hoped for. Your team decide to measure Mr Bailey's serum Clozapine level to see whether it falls within a therapeutic range. You might then be in a position to adjust the dose accordingly. Which one of the following statements is true:

Select one:

- A therapeutic range has been established for clozapine
- Dose responses for clozapine are not affected by genetic polymorphisms
- Nuclear Magnetic Resonance Spectroscopy is the most common method employed in therapeutic drug monitoring
- Clozapine has a wide therapeutic index
- The use of other psychotropic agents doesn't affect clozapine's therapeutic range

Your answer is incorrect.

Clozapine plasma levels can be useful in optimising treatment. In those not responding to clozapine, dose should be adjusted to give plasma levels in the range 350-600ug/L (a range reflecting a consensus in the literature). Plasma levels are generally lower in younger patients, males, and smokers, and higher in Asians. Much lower doses of clozapine are required in East Asians, Indians, and Bangladeshis. The prevalence of clozapine poor metabolisers is also higher in East Asians. Plasma levels do seem to predict EEG changes and seizures occur more frequently in patients with levels above 1000ug/L, so levels should be kept well below this. Other non-neurological clozapine-related adverse effects also seem to be plasma-level related, as might be expected. An upper limit of concentrations around 600-800ug/L has been proposed.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 838.

The correct answer is: A therapeutic range has been established for clozapine

Question 83

Not answered

Marked out of 1.00

Which of the following is most likely to be a feature of l-dopa withdrawal in a patient who has been treated for Parkinson's disease and associated dementia?

Select one:

- Apathy
- Mania
- Hyperprolactinemia
- Hallucinations
- Hyperreflexia

Your answer is incorrect.

Explanation:

After abrupt discontinuation of high-dose levodopa, a state of akinetic mutism can occur. Severe health risks may include stiffness, rigidity, tremor, and increased risk for thrombosis. When attempting to reduce levodopa dosage, a worsening of motor performance is common due to its short half-life. Similarly, possible worsening in dysphagia can occur in advanced stages of Parkinson disease. There are also several case reports of neuroleptic malignant syndrome after discontinuation of levodopa treatment. Changes in personality accentuation have also been documented, with a trend toward 'hypodopaminergic' behaviour with harm-avoiding tendencies, depression, pessimism, and apathy.

Ref: Koschel J et al. Implications of dopaminergic medication withdrawal in Parkinson's disease. Journal of Neural Transmission. 2021 Jul 29:1-0.

The correct answer is: Apathy

Question 84

Not answered

Marked out of 1.00

Raymond Williams is a 56-year-old man with a family history of schizophrenia and epilepsy. He had his first epileptic seizure as a teenager, and then in his early 20s developed a psychotic illness requiring treatment with an antipsychotic. At the time he was commenced on a medication, which increased his seizure frequency, and so it was stopped. He's since been on a number of different antipsychotic medications. Of the following, which is least likely to lower the seizure threshold?

Select one:

- Haloperidol
- Olanzapine
- Clozapine
- Sulpiride
- Chlorpromazine

Your answer is incorrect.

Of those listed, Haloperidol seems to be less associated with seizures (Ref: Drugs Today 2003(39);7:551-5517) Clozapine is particularly associated with increased risk of seizures.

The correct answer is: Haloperidol

Question 85

Not answered

Marked out of 1.00

Mr Andrew Cairns is a 70-year-old man who has presented to his GP with tiredness, muscular weakness and marked thirst. He is subsequently diagnosed with diabetes insipidus. Which of the following is most likely to be seen in his investigations?

Select one:

- High plasma osmolality, low urine osmolality
- Low plasma osmolality, low urine osmolality
- Low plasma osmolality, high urine osmolality
- High plasma osmolality, high urine osmolality
- High plasma osmolality, normal urine osmolality

Your answer is incorrect.

Explanation:

A high-normal plasma sodium concentration (>142 mmol/L) in conjunction with a urine osmolality that is lower than plasma osmolality points towards diabetes insipidus. A low plasma sodium concentration (<137 mmol/L) with a low urine osmolality (e.g. less than half the plasma osmolality) is usually indicative of water overload due to psychogenic polydipsia.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 269.

The correct answer is: High plasma osmolality, low urine osmolality

Question 86

Not answered

Marked out of 1.00

Ben Bradley is a 27-year-old man, who attends the outpatient clinic. He has a diagnosis of schizophrenia, and takes an antipsychotic (Haloperidol) to reduce his voice-hearing experience. After a two-year period of not smoking, he tells you that two weeks ago he has taken it up again, smoking about 15 cigarettes daily. You have in mind that his smoking may alter the effect of his antipsychotic. Which of these is true regarding smoking and metabolism?

Select one:

- It induces Cytochrome P450 (CYP)
- It enhances the analgesic effects of some opioids
- It reduces the clearance of heparin
- It inhibits Alanine Transaminase
- It enhances the sedative effects of benzodiazepines

Your answer is incorrect.

Explanation:

Tobacco smoke contains polycyclic aromatic hydrocarbons that induce (increase the activity of) certain hepatic enzymes (CYP1A2 in particular). The extent of enzyme induction is determined by the number and type of cigarettes smoked and the degree of smoke inhalation. For some drugs used in psychiatry, smoking significantly reduces drug plasma levels and higher doses are required than in non-smokers. Smoking may also affect alcohol metabolism by inducing CYP2E1.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 856.

The correct answer is: It induces Cytochrome P450 (CYP)

Question 87

Not answered

Marked out of 1.00

A drug should satisfy certain criteria to be suitable for therapeutic drug monitoring. Which of the following is a suitable criterion?

Select one:

- Narrow therapeutic index
- Unknown target concentration range
- Predominant hepatic metabolism
- Poor relationship between plasma concentrations and clinical effects
- Low pharmacokinetic variability

Your answer is incorrect.

Explanation:

Clear reasons for plasma level monitoring are:

1.To confirm compliance

2.If toxicity is suspected

3.If a pharmacokinetic drug interaction is suspected

4.If clinical response is difficult to assess directly and where a target range of plasma levels has been established

5.If the drug as a narrow therapeutic index and toxicity concerns are considerable

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 834.

The correct answer is: Narrow therapeutic index

Question 88

Not answered

Marked out of 1.00

Which of the following is a risk factor for clozapine-induced constipation?

Select one:

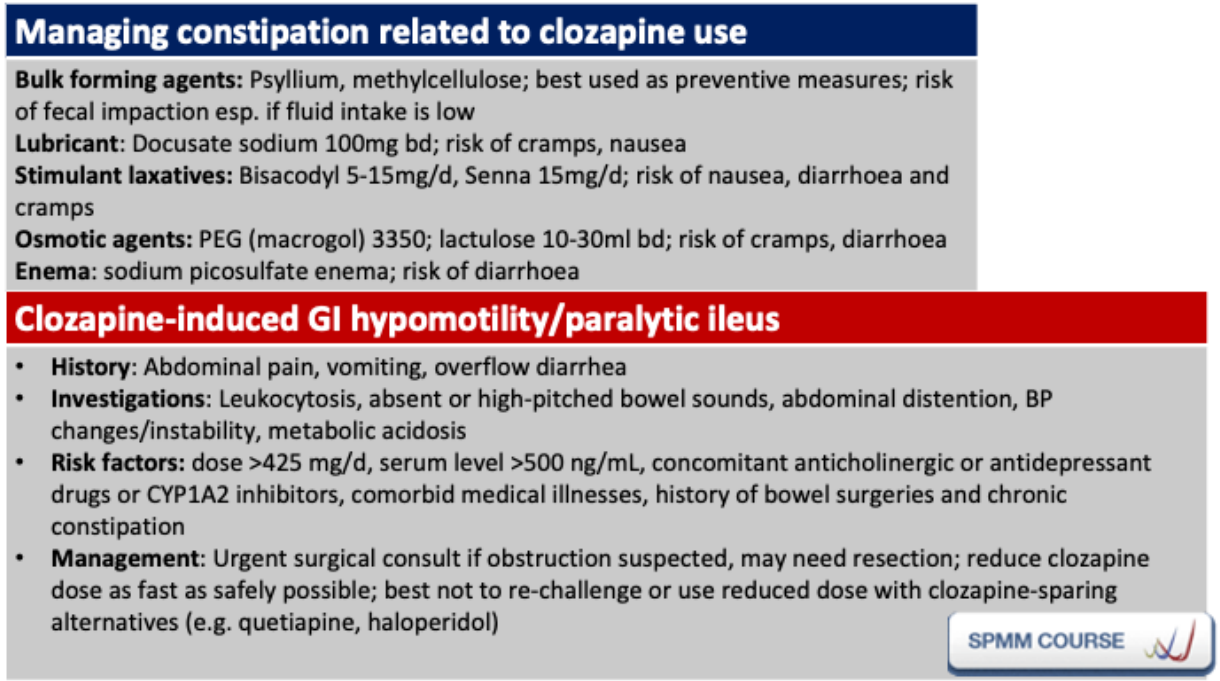
- Low body weight
- Female sex
- Low dose, administered once-daily
- Using liquid form of preparation
- Younger age

Your answer is incorrect.

Explanation:

Risk factors for developing clozapine-induced constipation include increasing age, female sex, anticholinergic medication, higher clozapine dose/plasma level, hypercalcaemia, gastrointestinal disease, obesity, diaphoresis, low-fibre diet, poor bowel habit, dehydration (exacerbated by hypersalivation), diabetes, hypothyroidism, Parkinson disease, multiple sclerosis.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 232.



The correct answer is: Female sex

Question 89

Not answered

Marked out of 1.00

Which of the following anticonvulsants is most likely associated with agitation and aggression as an adverse effect?

Select one:

- Topiramate
- Valproate
- Carbamazepine
- Gabapentin
- Zonisamide

Your answer is incorrect.

Explanation:

Based on available data, levetiracetam and topiramate seem to be associated with increased rates of irritability, hostility, and/or aggression, particularly in patients with a previous history of psychiatric symptoms. Anti-epileptic drugs with the highest incidence of disrupted behaviours are topiramate (2-10% have aggression related symptoms), levetiracetam (13% have non-psychotic behavioural symptoms) and vigabatrin (hyperactivity and restlessness). Psychosis is less common but most frequently seen with zonisamide and topiramate. There are reasonable data supporting no specific risk of aggression-related behaviour with carbamazepine, gabapentin, lamotrigine, and oxcarbazepine.

Ref: Brodie MJ et al. Epilepsy, antiepileptic drugs, and aggression: an evidence-based review. Pharmacological reviews. 2016 Jul 1;68(3):563-602.

The correct answer is: Topiramate

Question 90

Not answered

Marked out of 1.00

A 48 year-old woman with bipolar disorder presents for an ECG evaluation for a minor surgical assessment. The automated ECG read-out shows abnormal patterns. Lithium level is 0.9 mmol/L, with no history of overdose. Which of the following is the most likely ECG abnormality induced by lithium use?

Select one:

- Wenckebach type 1 block
- Supraventricular tachycardia
- Torsades de pointes
- Wellen's syndrome
- Wolff-Parkinson-White Syndrome

Your answer is incorrect.

Most ECG changes in lithium users are innocuous. Common ECG changes in patients taking lithium include T wave inversion - the most frequently reported ECG finding. Other findings include sinus node dysfunction, sinoatrial blocks, PR prolongation, QT prolongation/dispersion, and ventricular tachyarrhythmias. These ECG changes are more likely to occur with chronic lithium overdoses than acute overdoses. Sinoatrial block comprising reversible first and second degree (type I Wenckebach and type II) blocks are typically reported in patients on chronic lithium therapy for years. Wellens syndrome is also referred to as LAD coronary T-wave syndrome where as T-wave changes are associated with critical stenosis of the proximal left anterior descending (LAD) coronary artery. It is not a feature of lithium induced ECG changes.

The correct answer is: Wenckebach type 1 block

Question 91

Not answered

Marked out of 1.00

Dominic has a depressive disorder and willing to take medications. He is worried about sexual side effects. Which of the following antidepressants is associated with minimal sexual dysfunction?

Select one:

- Agomelatine
- Venlafaxine
- Duloxetine
- Fluovoxamine
- Paroxetine

Your answer is incorrect.

Explanation:

Lower risk antidepressants include agomelatine, bupropion, mirtazapine, vilazodone, vortioxetine, and moclobemide. Of these, agomelatine, bupropion, and vortioxetine have the best evidence supporting a more favourable sexual adverse-effect profile. Rates of sexual dysfunction with agomelatine may be similar to placebo.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 405-406.

The correct answer is: Agomelatine

Question 92

Not answered

Marked out of 1.00

Naima is a 45-year-old lady with schizophrenia. She is post-menopausal and not sexually active. Since her antipsychotic dose was increased she has noticed some milky discharge from her breasts, associated with a feeling of fullness. Her psychiatrist arranges some blood tests, which show a prolactin of 1245 mIU/L. She wants to switch Naima to a different antipsychotic. Which of the following drugs is least likely to be associated with hyperprolactinemia?

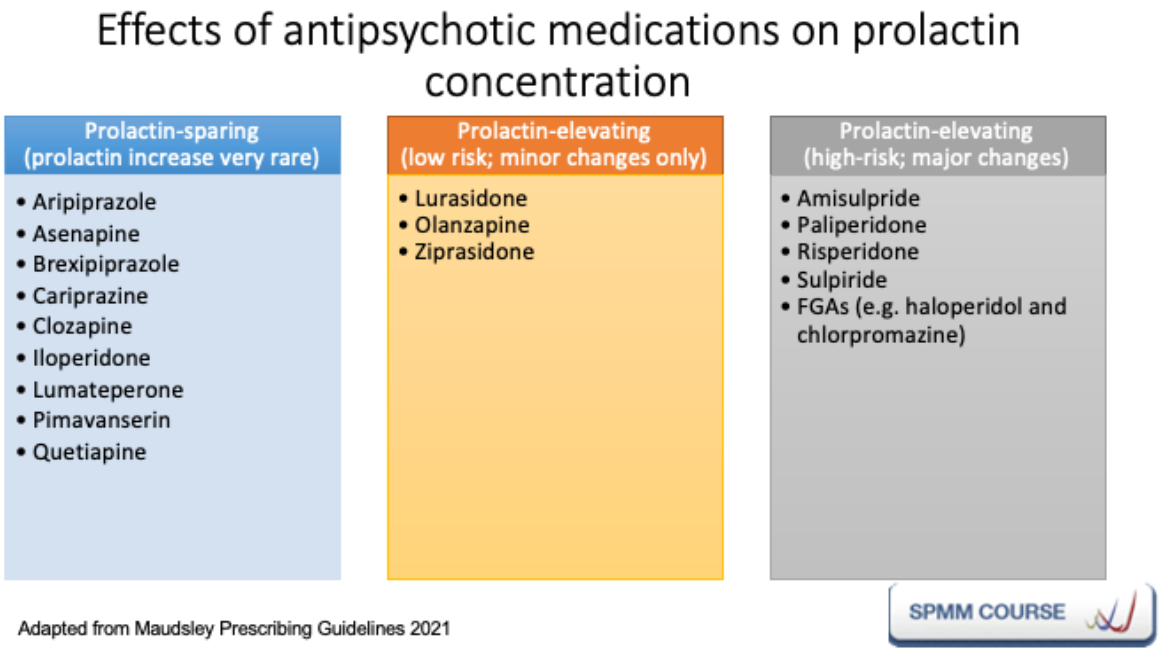
Select one:

- Aripiprazole
- Risperidone
- Amisulpride
- Haloperidol
- Sulpiride

Your answer is incorrect.

Explanation:

When D2 antagonists are administered, D2 receptors in the tuberoinfundibular dopamine pathway are blocked. This causes plasma prolactin concentrations to rise, a condition called hyperprolactinaemia. This can be associated with galactorrhea (i.e. breast secretions), sexual dysfunction, gynaecomastia (enlargement of the breasts) in men, and amenorrhea (i.e. irregular or lack of menstrual periods) in women.



Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 164-165.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 168.

The correct answer is: Aripiprazole

Question 93

Not answered

Marked out of 1.00

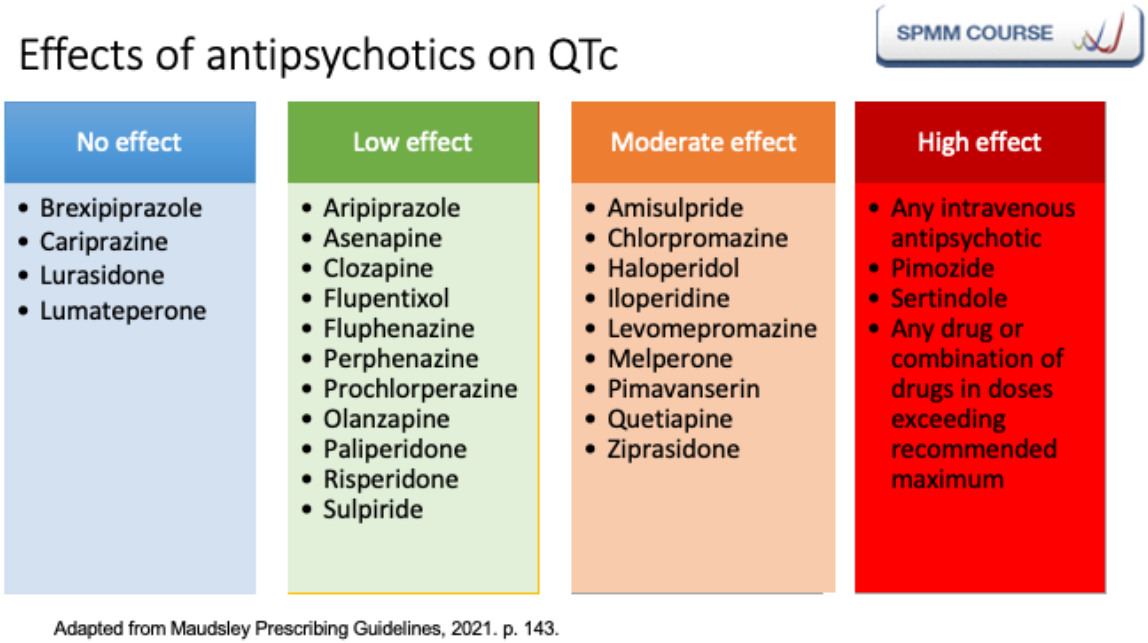
Miss Norman is currently being treated for Schizophrenia. She has a history of sudden cardiac death in her family and has read that some psychiatric medications can increase the risk of prolonged QTc and torsade de pointes. Which of the following antipsychotic is least likely to cause QTC prolongation?

Select one:

- Lurasidone
- Haloperidol
- Clozapine
- Olanzapine
- Sulpiride

Your answer is incorrect.

Explanation:



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 143.

The correct answer is: Lurasidone

Question 94

Not answered

Marked out of 1.00

Jennifer has been brought to A&E with a reduced conscious state after a suspected overdose of an unknown substance. Flumazenil is considered as a possible treatment. Regarding flumazenil, which is the correct mechanism of action:

Select one:

- Benzodiazepine antagonist
- Benzodiazepine partial agonist
- Opioid partial agonist
- Serotonin(5HT) 2A antagonist
- Muscarinic agonist

Your answer is incorrect.

Explanation:

Flumazenil is a benzodiazepine receptor antagonist. It blocks benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex, preventing benzodiazepines from binding there. It has a terminal half-life of 41-79 minutes. It may induce seizures, particularly in patients tolerant to or dependent on benzodiazepines, or who have overdosed on cyclic antidepressants, received recent/repeated doses of parenteral benzodiazepines, or have jerking or convulsion during overdose.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 293-294.

The correct answer is: Benzodiazepine antagonist

Question 95

Not answered

Marked out of 1.00

Which of the following acts as a messenger in the Reticular Activating System and contributes to sleep architecture?

Select one:

- Nitric oxide
- Vasopressin
- Nitrous oxide
- Anandamide
- Glycine

Your answer is incorrect.

Explanation:

Numerous lines of evidence have suggested a role for nitric oxide in the regulation of sleep-wake cycles. Neuronal nitric oxide synthase (nNOS)-expressing neurons occur in several areas that initiate REM sleep. In animal models, microinjection of compounds that release nitric oxide decreases wakefulness and increases slow-wave sleep compared with controls. Consistent with this, NOS inhibitors show a trend toward decreasing slow-wave and REM sleep.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 970.

The correct answer is: Nitric oxide

Question 96

Not answered

Marked out of 1.00

Wilfred Williamson is 86 years old. He's been experiencing progressive memory deficits for the past 5 years. His daughter has been reading up on the neurochemistry of dementia, and understands that the amino acid glycine (or D-serine) acts as a neurotransmitter, as a co-agonist alongside glutamate. At which of these receptors does glycine exert its action?

Select one:

- NMDA receptor
- Nicotinic Acetylcholine receptor
- GABA receptor
- Kainate receptor
- 5-HT3 receptor

Your answer is incorrect.

Explanation:

Glutamate systems are curious in that one of the key receptors for glutamate (the NMDA receptor) requires a co-transmitter in addition to glutamate in order to function. The co-transmitter is either the amino acid glycine or another amino acid, closely related to glycine, known as D-serine.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 97-98.

The correct answer is: NMDA receptor

Question 97

Not answered

Marked out of 1.00

Zolpidem is known to have a high affinity for which one of the following receptors?

Select one:

- Omega-1 receptor
- AMPA receptors
- Sigma-1 receptor
- Omega-2 receptor
- Sigma-2 receptor

Your answer is incorrect.

Explanation:

Z drugs bind to benzodiazepine-sensitive GABA-A receptors. A different nomenclature (omega system) denotes the specific sites of GABA-A receptors where these 'Z' drugs act as omega-1 receptors. Conventional benzodiazepines interact at both ω1 and ω2 sites, the latter likely contributing to the cognitive side effects o benzodiazepines.

Zolpidem and zaleplon bind selectively to alpha-1 subunits of these sites. By contrast, benzodiazepines and zopiclone/eszopiclone bind to four alpha subunits (alpha-1, alpha-2, alpha-3, and alpha-5). The functional significance of alpha-1 selectivity is not yet proven, but may contribute to lower risk of tolerance and dependence. The alpha-1 subtype is known to be critical for producing sedation and thus is targeted by every effective GABA-A PAM (positive allosteric modulator) hypnotic, both benzodiazepines and Z-drugs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 422-423.

Madari S, Golebiowski R, Mansukhani MP, Kolla BP. Pharmacological management of insomnia. Neurotherapeutics. 2021 Jan;18(1):44-52.

The correct answer is: Omega-1 receptor

Question 98

Not answered

Marked out of 1.00

Mr Parker is an 83-year-old man who has been struggling with symptoms of severe depression for the past six weeks. One week ago he stopped eating, became mute, and now has significantly reduced his fluid intake. You have a conversation with his family about the option of using ECT. Mr Parker's daughter, who is a laboratory scientist, is interested in the neurochemical effects of ECT. Which of the following statements about neurochemical findings associated with ECT is correct? There is:

Select one:

- Reduction in serotonin (5-HT2) receptor density
- Deactivation of dopaminergic neurotransmission
- Decreased concentration of brain-derived neurotrophic factor
- Increase in blood cortisol levels
- Decrease in cerebral acetylcholine

Your answer is incorrect.

Explanation:

ECT has an effect on almost all the major neurotransmitters in the brain such as serotonin, dopamine, acetylcholine, endogenous opioids, epinephrine and norepinephrine, etc. Similar to antidepressants, ECT down-regulates 5-HT2 receptors globally in the brain of patients with MDD. Reduction in 5-HT2 receptor density is suggested to have antidepressant action. ECT has been proposed to modulate dopaminergic receptors and enhance dopamine neurotransmission in the brain. Differential BDNF expression in different areas of the brain is believed to be responsible for antidepressant action of electroconvulsive seizures. ECT has also been shown to reduce the cortisol levels to normal among patients with MDD in due course.

Ref: Singh A, Kar SK. How Electroconvulsive Therapy Works?: Understanding the Neurobiological Mechanisms. Clinical Psychopharmacology and Neuroscience. 2017 Aug.

The correct answer is: Reduction in serotonin (5-HT2) receptor density

Question 99

Not answered

Marked out of 1.00

Pindolol is an antihypertensive medication that has also been used as an augmentation treatment in depression. Which of the following best describes its mechanism of action?

Select one:

- 5-HT1-A antagonist
- 5- HT1-B agonist
- 5-HT2-A agonist
- 5-HT2-A antagonist
- 5-HT1-B antagonist

Your answer is incorrect.

Explanation:

Pindolol is primarily an adrenergic beta-blocking drug, which also blocks serotonin 1A auto-receptors. The co-administration of pindolol with a serotonergic antidepressant could be expected to result in an immediate increase in serotonin neurotransmission, thus eliminating the delay in onset of antidepressant response. As well as being used to speed the onset of antidepressant response, pindolol has also been used to augment the efficacy of antidepressant drugs in acute-phase non-responders and treatment-resistant depression.

Ref: National Institute for Health and Clinical Excellence. The NICE guideline on the treatment and management of depression in adults. Updated edition 2019.

The correct answer is: 5-HT1-A antagonist

Question 100

Not answered

Marked out of 1.00

Medications used in Parkinson's disease work via a variety of mechanisms. Which of the following is used to facilitate CNS dopaminergic transmission through NMDA mediated actions?

Select one:

- Amantadine
- Levodopa
- Carbidopa
- Benserazide
- Selegiline

Your answer is incorrect.

Explanation:

Amantadine's mechanism of action is thought to be weak antagonism of NMDA (n-methyl-D-aspartate) glutamate receptors, possibly leading to downstream changes in the activity of dopamine in both the direct and indirect striatal motor pathways. Carbidopa and Benserazide is a peripherally acting DOPA decarboxylase inhibitor, which prevents the breakdown of levodopa and thus increases its availability in the CNS. Selegiline and Rasagiline are MAO-B inhibitors and reduce the breakdown of dopamine. Levodopa is a precursor of dopamine and can increase concentrations of this in the CNS when taken orally.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 169.

The correct answer is: Amantadine

Question 101

Not answered

Marked out of 1.00

Jane is a slim 24-year-old lady who has recently been diagnosed with schizophrenia and is under the care of the early intervention service. She often has low blood pressure and has heard that hypotension can occur with antipsychotics. She is discussing treatment options with you in the clinic. Which of the following antipsychotics is least likely to cause hypotension?

Select one:

- Aripiprazole
- Quetiapine
- Clozapine
- Olanzapine
- Chlorpromazine

Your answer is incorrect.

Explanation:

Antipsychotics with a high affinity for postsynaptic alpha-1-adrenergic receptors are most frequently implicated in postural hypotension. Among the SGAs, the reported incidence is highest with clozapine, quetiapine, and iloperidone. Lurasidone, asenapine, amisulpride, aripiprazole, brexipiprazole, and cariprazine are considered low risk for postural hypotension. There are limited quantitative data for FGAs, but low-potency phenothiazines (e.g. chlorpromazine) are considered most likely to cause orthostatic hypotension.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 161, 183.

The correct answer is: Aripiprazole

Question 102

Not answered

Marked out of 1.00

Propranolol is a commonly used anti-hypertensive, which has applications in a variety of other physical conditions and in managing performance anxiety. Which of the following best describes its mechanism of action?

Select one:

- Beta 1 and beta 2 adrenergic antagonist
- Beta 2 adrenergic antagonist
- Beta 2 and beta 3 adrenergic antagonist
- Beta 1 adrenergic antagonist
- Beta 1, beta 2 and beta 3 adrenergic antagonist

Your answer is incorrect.

Explanation:

Propranolol is a beta 1 and 2 adrenergic antagonist. When used to treat tremor, antagonism of peripheral beta 2 receptors is the proposed mechanism. When used in PTSD, blockade of beta-1 adrenergic receptors may theoretically prevent fear conditioning and reconsolidation of fear.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 653.

The correct answer is: Beta 1 and beta 2 adrenergic antagonist

Question 103

Not answered

Marked out of 1.00

Most antidepressants work by acting upon serotonin or noradrenaline functioning in the CNS. Which of the following antidepressants works upon dopamine activity in the brain?

Select one:

- Bupropion
- Vortioxetine
- Reboxetine
- Agomelatine
- Mianserin

Your answer is incorrect.

Explanation:

Bupropion is unique among antidepressants. It is a noradrenergic and dopaminergic reuptake inhibitor. Bupropion is indicated for depression but is only licensed in the UK for treatment of nicotine dependence.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 290.

The correct answer is: Bupropion

Question 104

Not answered

Marked out of 1.00

Which of the following psychotropic drugs could be potentially affected by acetylator status?

Select one:

- Phenothiazines
- Lithium
- Tricyclic antidepressants
- Acetylcholinesterase inhibitor
- MAO inhibitors

Your answer is incorrect.

Though potentially MAO inhibitors can be affected acetylator status, in practical use this is not usually a problem.

The correct answer is: MAO inhibitors

Question 105

Not answered

Marked out of 1.00

Which one among the following antidementia drugs is mainly renally excreted?

Select one:

- Tacrine
- Donepezil
- Rivastigmine
- Galantamine
- Memantine

Your answer is incorrect.

Explanation:

Memantine undergoes little metabolism and is mostly excreted unchanged in the urine. No dose adjustment is needed in mild or moderate renal impairment. Dose reduction is recommended in severe renal impairment.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 474-475.

The correct answer is: Memantine

Question 106

Not answered

Marked out of 1.00

Which drug acts as a full agonist at Mu receptor and has a long half life?

Select one:

- Buspirone
- Buprenorphine
- Disulfiram
- Methadone
- Acamprosate

Your answer is incorrect.

Explanation:

Methadone is a full agonist at mu opioid receptors and has a long half-life (24 hours).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 478.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 636.

The correct answer is: Methadone

Question 107

Not answered

Marked out of 1.00

When swapping MAOIs to SSRIs, what is the optimum wash out period?

Select one:

- Minimum 2 days is essential
- Minimum 5 days is essential
- Minimum 2 months is essential
- Minimum 2 weeks is essential
- Minimum 5 weeks is essential

Your answer is incorrect.

MAOI to SSRI- 2 weeks SSRI to MAOI-2 weeks for all, 5 weeks for Fluoxetine TCA to MAOI- 2 weeks MAOI to TCA- 1-2 weeks.

The correct answer is: Minimum 2 weeks is essential

Question 108

Not answered

Marked out of 1.00

Apparent volume of distribution of haloperidol is

Select one:

- 1l/kg
- 0.5l/kg
- 0.1l/kg
- 0.25l/kg
- more than 2l/kg

Your answer is incorrect.

Explanation:

The apparent volume of distribution is the theoretical volume of fluid into which the total drug administered would have to be diluted to produce the concentration in plasma. For example, if 1000 mg of a drug is given and the subsequent plasma concentration is 10 mg/L, that 1000 mg seems to be distributed in 100 L (dose/volume = concentration; 1000 mg/x L = 10 mg/L; therefore, x = 1000 mg/10 mg/L = 100 L). Many acidic drugs (e.g., warfarin) are highly protein-bound and thus have a small apparent volume of distribution. Many basic drugs (e.g., amphetamine) are extensively taken up by tissues and thus have an apparent volume of distribution larger than the volume of the entire body. Haloperidol is very extensively bound to plasma proteins (90%). The apparent volume of distribution of haloperidol has been found to range from 9.5-21.7 L/kg. This high volume of distribution is in accordance with its lipophilicity, which also suggests free movement through various tissues, including the blood-brain barrier.

Ref: https://go.drugbank.com/drugs/DB00502

The correct answer is: more than 2l/kg

Question 109

Not answered

Marked out of 1.00

Among Caucasians who are social drinkers, what percentage of alcohol is metabolized by alcohol dehydrogenase?

Select one:

- less than 9
- 60-79
- more than 90%
- 75-85
- 40-59

Your answer is incorrect.

The major site of alcohol metabolism or breakdown (biotransformation) is in the liver. Alcohol dehydrogenase is the rate-limiting enzyme in the metabolism of alcohol and thus limits the amount of alcohol metabolized. More than 90% of consumed alcohol in Caucasians is metabolized via this route. Another significant metabolic pathway in the liver is called microsomal ethanol oxidizing system (MEOS). This consists of the induction of the liver enzyme cytochrome P-450 (CYP2E1) which biotransforms a number of substances including alcohol and barbiturates

The correct answer is: more than 90%

Question 110

Not answered

Marked out of 1.00

Which of the following over-the-counter preparations must be avoided by a patient who is prescribed phenelzine for resistant depression with social anxiety?

Select one:

- Nasal decongestant sprays
- Artificial tears
- Antacids
- Nicotine inhalers
- Heat patches

Your answer is incorrect.

Explanation:

Over-the-counter medications to avoid taking an MAOI such as phenelzine or tranylcypromine include cold and cough preparations, including those containing dextromethorphan, decongestants (e.g. phenylephrine, pseudoephedrine), hay fever medication, sinus medications, asthma inhalant medications, weight reducing preparations, and 'pep' pills. If MAOIs are taken with decongestants, the patient may experience a life-threatening rise in blood pressure due to synergistic sympathetic drive.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 615.

The correct answer is: Nasal decongestant sprays

Question 111

Not answered

Marked out of 1.00

Which of the following statements about lithium is correct?

Select one:

- Lithium levels fall within 10 days of starting a thiazide diuretic
- NSAIDs can increase the plasma levels of lithium to >400%
- A third of patients treated with lithium experience at least one side effect
- 70% of middle aged women develop hypothyroidism after 6-12 months of treatment
- ACE inhibitors cause a reduction in lithium levels

Your answer is incorrect.

Explanation:

NSAIDs inhibit the synthesis of renal prostaglandins, thereby reducing renal blood flow and possibly increasing renal re-absorption of sodium and therefore lithium. The magnitude of the rise is unpredictable for any given patient; case reports vary from increases of around 10% to over 400%. The levels of lithium usually increase within ten days of a thiazide diuretic being prescribed. ACE inhibitors can reduce thirst, which can lead to dehydration, and increase renal sodium loss, leading to increased sodium reabsorption by the kidneys, causing an increase in lithium plasma levels. About 20% of middle-aged women develop hypothyroidism after 6-12 months of treatment. Approximately two-thirds of patients treated with lithium experience at least one side effect.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 253.

The correct answer is: NSAIDs can increase the plasma levels of lithium to >400%

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